

Dopamine activity encodes the changing valence of the same stimulus in conditioned taste aversion paradigms

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eLife Assessment

This study utilizes an elegant approach to examine valence encoding of the mesolimbic dopamine system. The findings are **valuable**, demonstrating differential responses of dopamine to the same taste stimulus according to its valence (i.e., appetitive or aversive) and in alignment with distinct behavioral responses. The evidence supporting the claims is **convincing**, resulting from a well-controlled experimental design with minimal confounds and thorough reporting of the data.

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Abstract

Mesolimbic dopamine encoding of non-contingent rewards and reward-predictive cues has been well established. Considerable debate remains over how mesolimbic dopamine responds to aversion and in the context of aversive conditioning. Inconsistencies may arise from the use of aversive stimuli that are transduced along different neural paths relative to reward or the conflation of responses to avoidance and aversion. Here, we made intraoral infusions of sucrose and measured how dopamine and behavioral responses varied to the changing valence of sucrose. Pairing intraoral sucrose with malaise via injection of lithium chloride (LiCl) caused the development of a conditioned taste aversion (CTA), which rendered the typically rewarding taste of sucrose aversive upon subsequent re-exposure. Following CTA formation, intraoral sucrose suppressed the activity of ventral tegmental area dopamine neurons (VTA_{DA}) and nucleus accumbens (NAc) dopamine release. This pattern of dopamine signaling after CTA is similar to intraoral infusions of innately aversive quinine and contrasts with that to sucrose when it was novel or not paired with LiCl. Dopamine responses were negatively correlated with behavioral reactivity to intraoral sucrose and predicted home cage sucrose preference. Further, dopamine responses scaled with the strength of the CTA, which was increased by repeated LiCl pairings and weakened through extinction. Thus, the findings

demonstrate differential dopamine encoding of the same taste stimulus according to its valence, which is aligned to distinct behavioral responses.

Introduction

Dopamine neurons of the ventral tegmental area (VTA_{DA}) and their release of dopamine in the nucleus accumbens (NAc) play key roles in the encoding of primary reward (Cohen et al., 2012; Roitman et al., 2008; Romo and Schultz, 1989), reward-related learning (Schultz, 2016; Schultz et al., 1997; Watabe-Uchida et al., 2017) and motivation (Berridge and Robinson, 1998; Bromberg-Martin et al., 2010; Wise, 2004). Brief, phasic dopamine responses are evoked by primary rewarding stimuli and develop to their predictors (Konanur et al., 2023, for example). Pauses in dopamine activity are evoked by reward omission (Sugam et al., 2012; Tobler et al., 2003). Collectively, these findings support a role for dopamine in signaling reward prediction errors (RPEs) (Schultz, 2016, 1998; Schultz et al., 1997; Steinberg et al., 2013). This value-learning account has been more recently challenged to include the learning from, and updating of, the value of reward-directed actions (Coddington et al., 2023; Jeong et al., 2022). Yet there is little debate that dopamine increases to reward and reward-related cues. A role for dopamine in aversion and aversive conditioning is more controversial (Morales and Margolis, 2017). Some recordings of VTA_{DA} activity (Bromberg-Martin et al., 2010; Matsumoto and Hikosaka, 2009) or dopamine release (Mikhailova et al., 2019; Wenzel et al., 2015) support increased signaling in aversion and aversive conditioning while others support decreased signaling (Mileyskiy and Morales, 2011; Roitman et al., 2008; Ungless et al., 2004; Wheeler et al., 2011; Zhuo et al., 2024; see Morales and Margolis, 2017 for review).

The unsettled nature of dopamine signaling in aversion could be due to several factors. First, like reward (Berridge and Robinson, 2003), aversion is a multi-dimensional construct that ranges from involuntary responses to noxious primary stimuli to passive or active avoidance in response to predictors of aversion. Aversive stimuli have included foot/tail shocks (de Jong et al., 2019; Mileyskiy and Morales, 2011), air puffs (Matsumoto and Hikosaka, 2009; Mirenowicz and Schultz, 1996; Zhuo et al., 2024), painful pinches (Romo and Schultz, 1989), and white noise (Goedhoop et al., 2022). These are transduced along different sensory pathways (i.e., pain, somatosensory, auditory) from reward (typically gustatory; e.g. sucrose). Even when administering rewarding and aversive stimuli that utilize the same sensory pathway, as with taste, rewarding sweet and aversive bitter solutions activate different taste receptors (Schier and Spector, 2019). Complicating the picture further, mesolimbic dopamine responses to noxious stimuli have been assayed in anesthetized animals (Brischoux et al., 2009; Budygin et al., 2012; Romo and Schultz, 1989) – making behavioral assessment of aversion impossible. In awake, behaving recordings, behavioral paradigms often permit the avoidance of aversive stimuli (Kutlu et al., 2021; Oleson et al., 2012) which can ultimately be rewarding. To facilitate the sampling of aversive stimuli, water or food deprivation protocols are often used (Glover et al., 2016; Gordon-Fennell et al., 2023; Hurley et al., 2023; López et al., 2023; Miranda et al., 2023). However, dopamine responses are modulated by physiological state (i.e., hunger, thirst, sodium appetite (Cone et al., 2015; Fortin and Roitman, 2018; Hsu et al., 2020)), presenting a further confound in interpreting mesolimbic responses.

To overcome these limitations, we measured VTA_{DA} neural activity and dopamine release in the lateral shell of the NAc using fiber photometry as we varied the valence of intraoral sucrose infusions. In Paired rats, the valence of sucrose was changed by subsequently administering lithium chloride (LiCl) – which induces visceral malaise (Bernstein et al., 1992) and reliably conditions a taste aversion (CTA; Garcia et al., 1955; Garcia and Kimeldorf, 1957; Nachman and Ashe, 1973; Nolan et al., 1997; Kim et al., 2010; Swank and Bernstein, 1994; Thiele et al., 1996). In Unpaired rats, sucrose valence was unchanged by injecting LiCl 24hr after intraoral sucrose (Smith and Roll, 1967). Changes in valence were strengthened through multiple pairings

of sucrose and LiCl or weakened through extinction. Throughout, rats were fed and watered ad libitum to avoid physiological need as a confound. To assay the stereotypical appetitive and aversive behavioral responses to taste stimuli (Breslin et al., 1992 [↗](#); Grill and Norgren, 1978a [↗](#)), we used a deep learning algorithm (DeepLabCut, (Mathis et al., 2018 [↗](#))) and home cage sucrose preference task to assess the valence of intraoral sucrose. Across testing, VTA_{DA} activity and dopamine release in the NAc differentially responded to sucrose taste based on its affective value ascertained from behavioral reactivity and sucrose preference. Thus, we conclude that the mesolimbic dopamine system differentially encodes valence – reward versus aversion – and flexibly for the same stimulus.

Results

NAc dopamine differentially responds to primary taste stimuli and correlates with differential behavioral reactivity

We expressed the dopamine fluorescent sensor AAV1.Syn.Flex.GRAB_DA2h (GRAB_DA2h) in the lateral shell of the NAc and recorded real-time fluorescence via an indwelling fiber optic (**Figure 1A-B** [↗](#), **Supplementary Figure 1** [↗](#)). To characterize dopamine and behavioral responses to innately appetitive and aversive taste stimuli, we made intraoral infusions of sucrose and quinine in naïve rats. On average, intraoral infusions of sucrose and quinine evoked different dopamine release responses ($t(5) = 3.61$, $p < 0.05$; **Fig. 1C** [↗](#)), which is consistent with prior work from our group (Hsu et al., 2020 [↗](#); Roitman et al., 2008 [↗](#)). Across individual trials, responses to sucrose infusions were higher than those to quinine (**Fig. 1D** [↗](#)). Using Receiver Operating Characteristic (ROC) analysis (Cone et al., 2015 [↗](#); Green and Swets, 1974 [↗](#)), we found dopamine responses to sucrose and quinine delivery on individual trials to be highly discriminable (area under the ROC curve (AUC_{ROC})=0.81; **Fig. 1E** [↗](#)).

Appetitive and aversive tastes arouse well-characterized, stereotypical responses in rats (Grill and Norgren, 1978a [↗](#)). Specifically, intraoral infusions of appetitive taste stimuli evoke mouth movements and tongue protrusions, but the head and body are relatively still. In contrast, aversive taste stimuli cause gapes, headshakes, forelimb flails, and chin rubs (Grill and Norgren, 1978a [↗](#)) – and therefore greater head and whole-body movement. We hypothesized that these responses could be well captured by measuring movement of the nose and forepaws (**Fig. 2A** [↗](#)). Here, primary taste stimuli produced distinct nose and forepaw responses to intraoral infusions relative to a 5s baseline period just before infusion onset (behavioral reactivity, **Fig. 2B** [↗](#)). Quinine produced greater average nose ($t(5) = 3.69$, $p < 0.05$, **Fig. 2C** [↗](#)); and forepaw ($t(5) = 3.82$, $p < 0.05$; **Fig. 2D** [↗](#)) movement relative to sucrose in the same rats. Importantly, these behavioral measures were negatively correlated to dopamine responses from the same rats where greater dopamine responses were associated with less movement evoked by the stimulus (nose movement to dopamine: $r^2=0.44$, slope=-0.041; $p < 0.05$; forepaw moment to dopamine: $r^2=0.44$, slope =-0.029; $p < 0.05$; **Fig. 2E** [↗](#) & 2F).

Dopamine responses differentially encode the same taste stimulus based on behavioral reactivity

CTA robustly shifts behavior to the same taste stimulus from ingestion to aversion (Grill and Norgren, 1978b [↗](#)). Here, we measured either dopamine release or VTA_{DA} activity (**Supplementary Fig. 2** [↗](#)) and behavioral responses from initially naïve rats (Conditioning Day, CD) to intraoral infusions of 0.3M sucrose. We then administered either Saline (Unpaired) or malaise-inducing LiCl (Paired). The subsequent day, rats did not receive intraoral infusions but did receive the counterbalanced injection in their home cage and were untreated the following day (**Fig. 3A** [↗](#)). All rats then received another session of sucrose intraoral infusions (Test Day, TD).

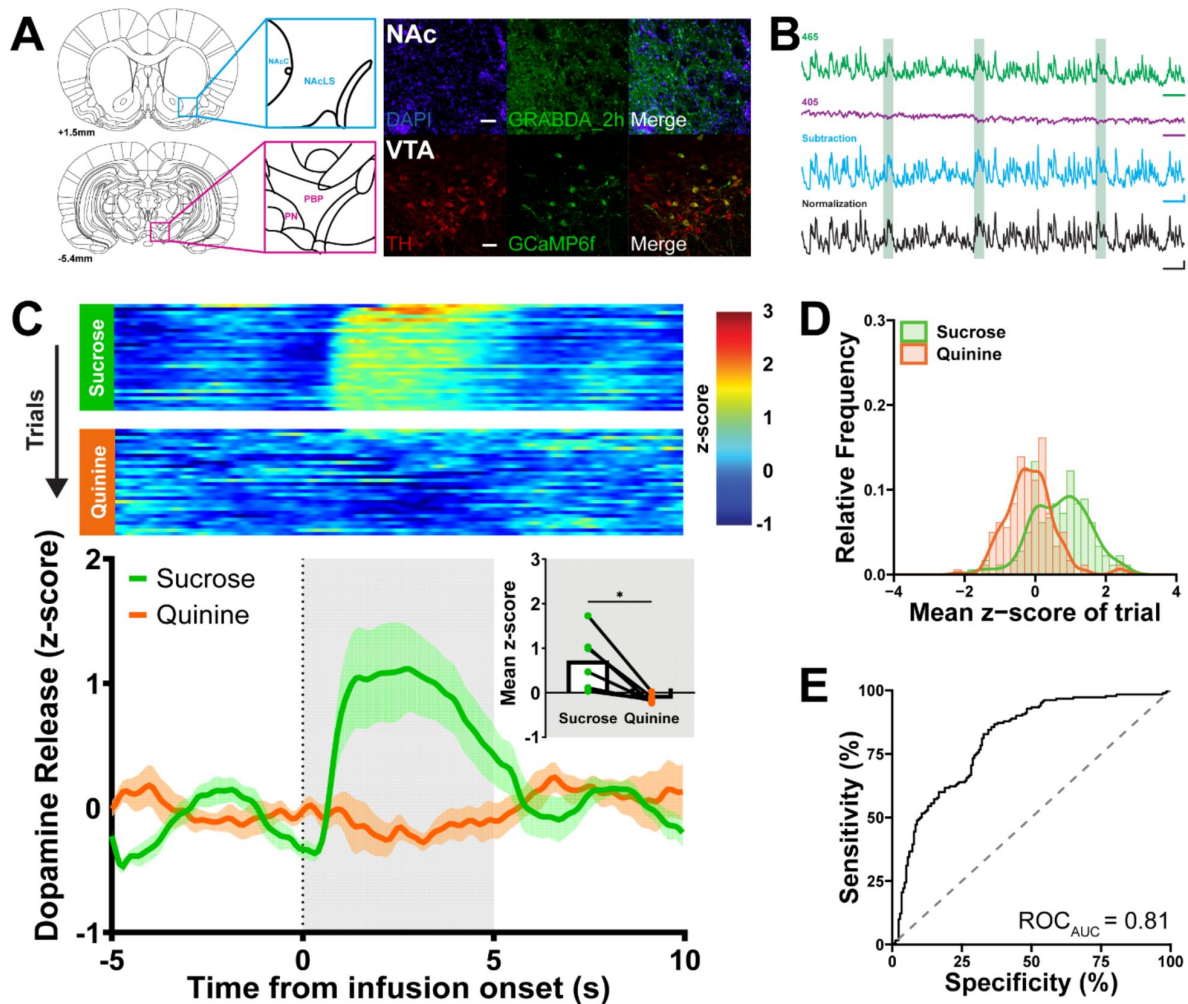


Figure 1.

In vivo fiber photometry in the mesolimbic dopamine system captures phasic dopamine responses to primary taste stimuli.

(A) Representative images of GRABDA_2h and Cre-dependent GCaMP6f expression in the NAc and VTA, respectively. *Top Row*: Dopamine release recordings from the lateral subregion of the NAc shell are confirmed via viral expression of dopamine-sensor, GRABDA2h (green), probed against DAPI (blue) to visualize sensor location targeted to the lateral shell of the NAc (NAcLS), which borders the NAc core (NAcC). *Bottom row*: Dopamine cellular activity was recorded in TH Cre+ rats in the VTA (paranigral nucleus (PN), the parabrachial pigmented area; (PBP)). TH+ (red) colocalized with intracellular calcium-sensor, GCaMP6f (green) to demonstrate isolation of the VTA_{DA} population. (B) Real-time dopamine release in the NAc across processing steps from a representative rat receiving 5s 0.3M sucrose intraoral infusions (light green bars). Fluorescence excited by the 465nm (Ca^{2+} - and GRABDA2h-dependent, green), and 405nm LED (Ca^{2+} -independent, purple) was captured. 465 and 405nm traces were scaled and subtracted to remove motion artifacts and photobleaching (blue). Fluorescence was then normalized to the whole recording session and represented as a z-score (black). (C) *Top*: heat maps for NAc dopamine release for 30 trials/session in which 200 μL of 0.3M sucrose (green) or 0.001M quinine (orange) was delivered over 5s. *Bottom*: Average dopamine release aligned to the onset of intraoral delivery. Dotted line represents onset of infusion and gray shading reflects infusion duration and time window for statistical analysis. *Bottom inset*: Average z-score during infusion period. (D) Relative frequency histogram of dopamine responses (mean z-score) to sucrose and quinine from every trial reported. (E) ROC of D determined a discriminable difference between dopamine responses. Dark lines and shading in C are represented as means; * $p < 0.05$ in paired t-test.

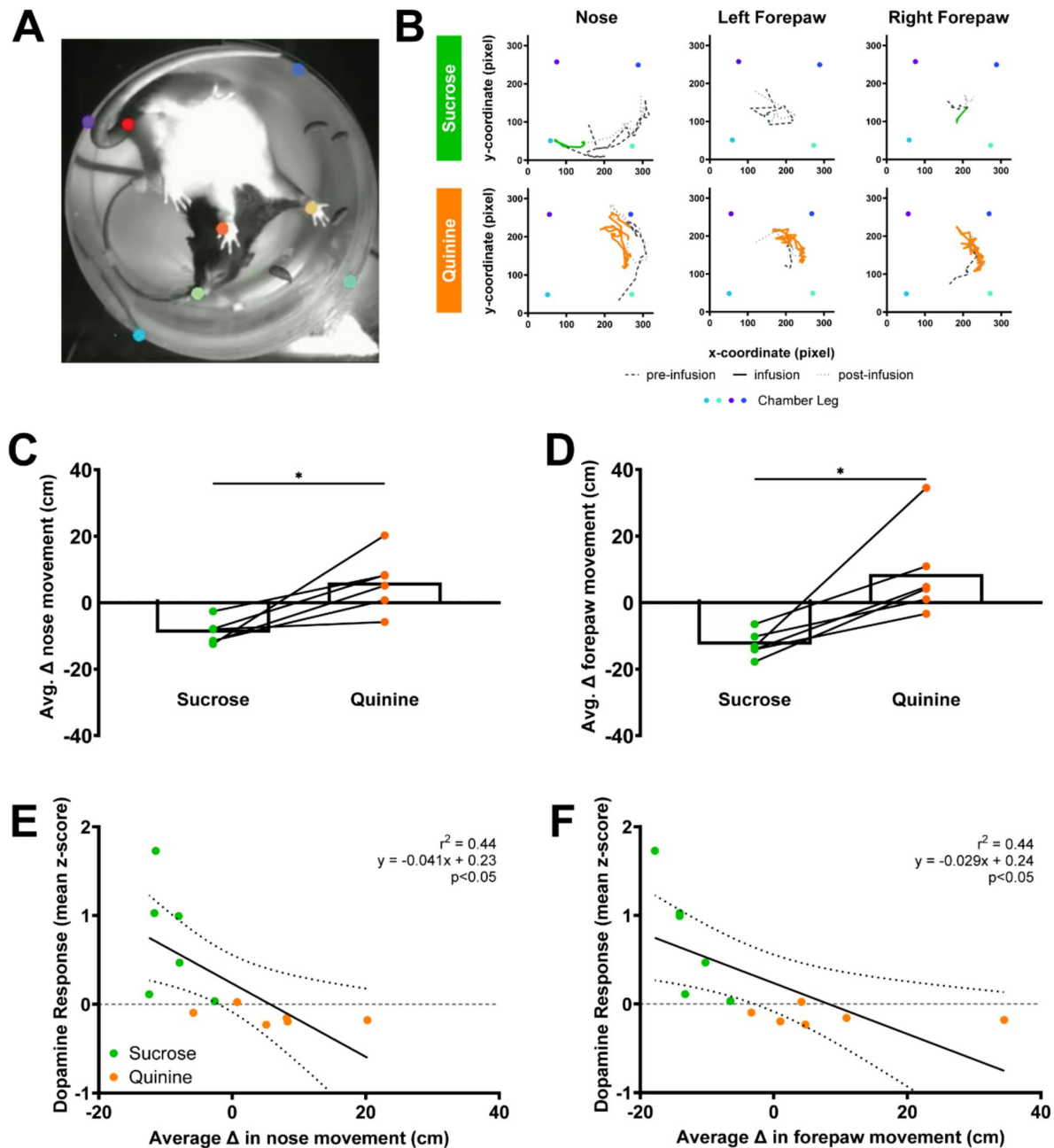


Figure 2.

Aversive taste stimuli are linked to increased movement of nose and forepaws.

(A) Representative image of nose, forepaws, tail base, and chamber legs tracking of a rat in a cylindrical chamber from a below chamber perspective. Positional coordinates of selected features were obtained using model created via DeepLabCut, an open-source deep-learning pose estimation program. Custom MATLAB scripts were used to analyze movement from the positional data. (B) Representative movement of nose and forepaws tracked by a DeepLabCut model during 5s pre-infusion, sucrose or quinine infusion, and post-infusion periods. (C&D) Behavioral reactivity was measured as the average change in nose movement or forepaw movement from baseline to infusion period. Intraoral infusion of quinine produces a greater behavioral reactivity. (E&F) Relationship between average change in behavioral reactivity and mean z-score of NAc dopamine during infusion averaged by session. Data in C & D are represented as means; * $p < 0.05$ in paired t-tests. Line in E & F denote the linear relationship between parameters with dotted lines as 95% confidence intervals. P-value of linear regressions indicate slope's deviation from zero.

Thus, by TD, all rats had equal exposure to all stimuli with the lone difference being that Paired rats had sucrose and LiCl administered in close temporal proximity whereas Unpaired rats had sucrose and LiCl administered at least 24hr apart. On TD, we measured each subjects' dopamine and behavioral responses to intraoral sucrose as per CD (**Fig. 3A**). In Unpaired rats, dopamine release to intraoral sucrose was unchanged between days (Unpaired: $t(9) = 0.83$, $p > 0.05$; **Figure 3B**). In sharp contrast, in Paired rats NAc dopamine release evoked by intraoral sucrose was significantly suppressed on TD relative to CD (Paired: $t(10) = 3.97$, $p < 0.01$; **Fig. 3B**). Using an ROC to test for between subject differences, we found that dopamine release to intraoral sucrose on individual trials from Paired and Unpaired rats could not be discriminated on CD ($AUC_{ROC} = 0.50$) but could be well discriminated on TD ($AUC_{ROC} = 0.80$; **Fig. 3C-F**).

Similar to dopamine release in the NAc, VTA_{DA} activity was unchanged between days in Unpaired rats (Unpaired: $t(5) = 0.60$, $p > 0.05$; **Fig 3G**) but suppressed on TD relative to CD in Paired rats (Paired: $t(6) = 2.627$, $p < 0.05$; **Fig 3G**). Likewise, individual VTA_{DA} responses between groups could not be discriminated on CD ($AUC_{ROC} = 0.55$) but were well discriminated on TD ($AUC_{ROC} = 0.76$; **Fig. 3H-K**).

We analyzed behavioral reactivity to sucrose infusions on CD and TD in Unpaired and Paired rats during which dopamine release or VTA_{DA} activity were also measured. For rats in which NAc dopamine release was measured, intraoral infusions of sucrose elicited different nose movement responses across testing days (Treatment (Unpaired vs Paired) $\times$ Day (CD vs TD): $F_{(1,18)} = 6.61$, $p < 0.05$; **Fig. 4A**). In the Unpaired rats, nose movement reactions were comparable between days ($p > 0.05$; **Fig. 4A**) but Paired subjects showed increased nose movement to intraoral infusions from TD relative to CD ($p = 0.0009$; **Fig. 4A**). Analysis of forepaw movement also indicated differences between Treatment groups (main effect of Treatment: $F_{(1,18)} = 8.81$, $p < 0.01$; **Fig. 4B**). Forepaw movement did not change from CD to TD in Unpaired rats ($p > 0.05$; **Fig. 4B**) but increased in Paired rats ($p < 0.05$; **Fig. 4B**). Further, Paired rats showed greater forepaw reactivity than Unpaired rats on TD ($p < 0.01$; **Fig. 4B**). Examining the relationship between dopamine release and behavioral reactivity revealed negative correlations between dopamine release and nose movement ($r^2 = 0.26$, slope $= -0.025$, $p < 0.001$; **Fig. 4C**) and between dopamine release and forepaw moment ($r^2 = 0.22$, slope $= -0.021$, $p < 0.005$; **Fig. 4D**).

Similar results were observed in the separate group of rats from which VTA_{DA} activity recordings were made. Intraoral infusions of sucrose elicited different nose movement responses between subjects across testing days (Treatment $\times$ Day: $F_{(1,10)} = 6.60$, $p < 0.05$; **Fig. 4E**). Specifically, nose movement was comparable between CD and TD in Unpaired rats ($p > 0.05$; **Fig. 4E**) but significantly greater on TD relative to CD in Paired rats ($p < 0.005$; **Fig. 4E**). Forepaw movement responses between subjects and across testing days were similar (Treatment $\times$ Day: $F_{(1,10)} = 6.34$, $p < 0.05$; **Fig. 4F**). Forepaw movement did not change across testing days in Unpaired rats ($p > 0.05$; **Fig. 4F**) but increased in Paired rats ($p < 0.005$; **Fig. 4F**). Further, dopamine activity was also negatively correlated to nose movement ($r^2 = 0.48$, slope $= -0.057$, $p < 0.0005$; **Fig. 4G**) and forepaw moment ($r^2 = 0.51$, slope $= -0.044$, $p < 0.0001$; **Fig. 4H**).

Repeated 'safe' re-exposure to sucrose reduces the suppression of sucrose-evoked phasic VTA_{DA} activity in parallel with behavioral extinction

A CTA can be extinguished when the conditioned stimulus (CS (i.e., intraoral sucrose)) is repeatedly presented in the absence of the unconditioned stimulus (US (i.e., malaise induced by LiCl injection) (Hadamitzky et al., 2015)); however, it is unclear if CTA-suppressed dopamine responses to intraoral sucrose recover with extinction. Here, VTA_{DA} activity was recorded (**Supplementary Fig. 3**) in a separate cohort of subjects that underwent Single-pairing CTA followed by five consecutive Extinction sessions, where intraoral sucrose was presented without

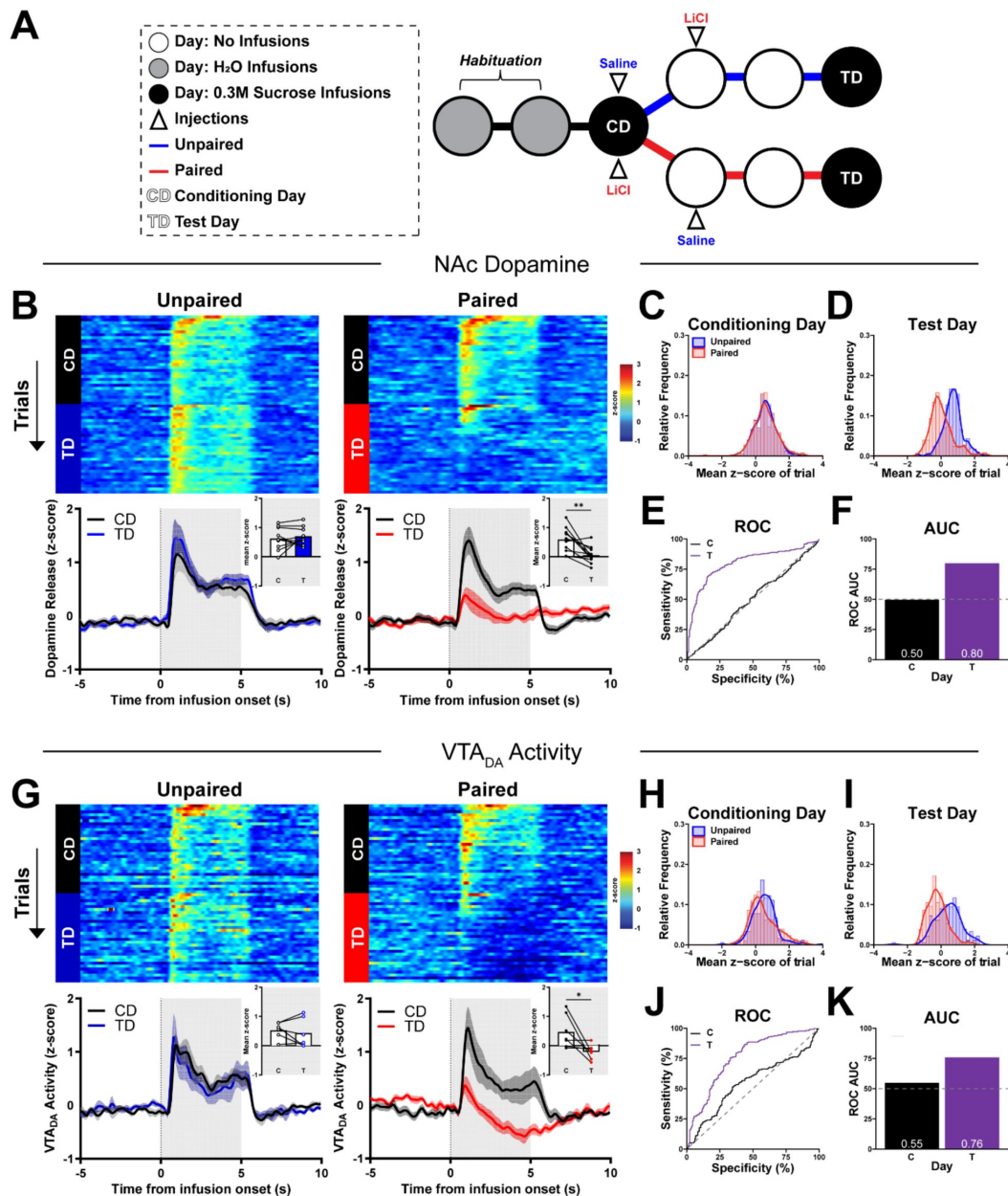


Figure 3.

Pairing of LiCl-induced malaise to sucrose suppresses phasic dopamine responses to intraoral sucrose delivery.

(A) Schematic of Single-pairing CTA timeline. Subjects were first habituated to 30 brief intraoral infusions (200 μ L/trial) of water at varying intertrial intervals (35-55s) daily for two days. On Conditioning Day (CD), rats received intraoral infusions of 0.3M sucrose and were then injected i.p. with saline (Unpaired) or 0.15M LiCl (Paired). Rats received the counterbalanced injection in the home cage the next day and were untreated the following day. On Test Day (TD), rats received intraoral infusions (parameters identical to CD). **(B)** NAc dopamine across trials and sessions before and after intraoral sucrose delivery onset. *Top*: NAc dopamine release (in z-score) across 30 trials/session during both CD and TD. *Bottom*: Average traces of dopamine release aligned to the onset of intraoral delivery of sucrose on CD and TD. *Insets*: Mean z-score quantified during 5s intraoral delivery of sucrose. **(C-D)** Relative frequency histogram of dopamine release responses to sucrose on CD and TD for every trial reported as mean z-score for both Unpaired and Paired subjects. **(E)** ROC of relative frequency distributions of mean z-score acquired from each trial of on CD and TD between treatment groups. **(F)** Plotted area under the curve (AUC_{ROC}) values of **E**. **(G-K)** Recordings of VTA_{DA} activity from Unpaired and Paired rats reported with same conventions as **B-F**. Data in **B & G** are represented as means; * $p < 0.05$, ** $p < 0.01$, paired t-test.

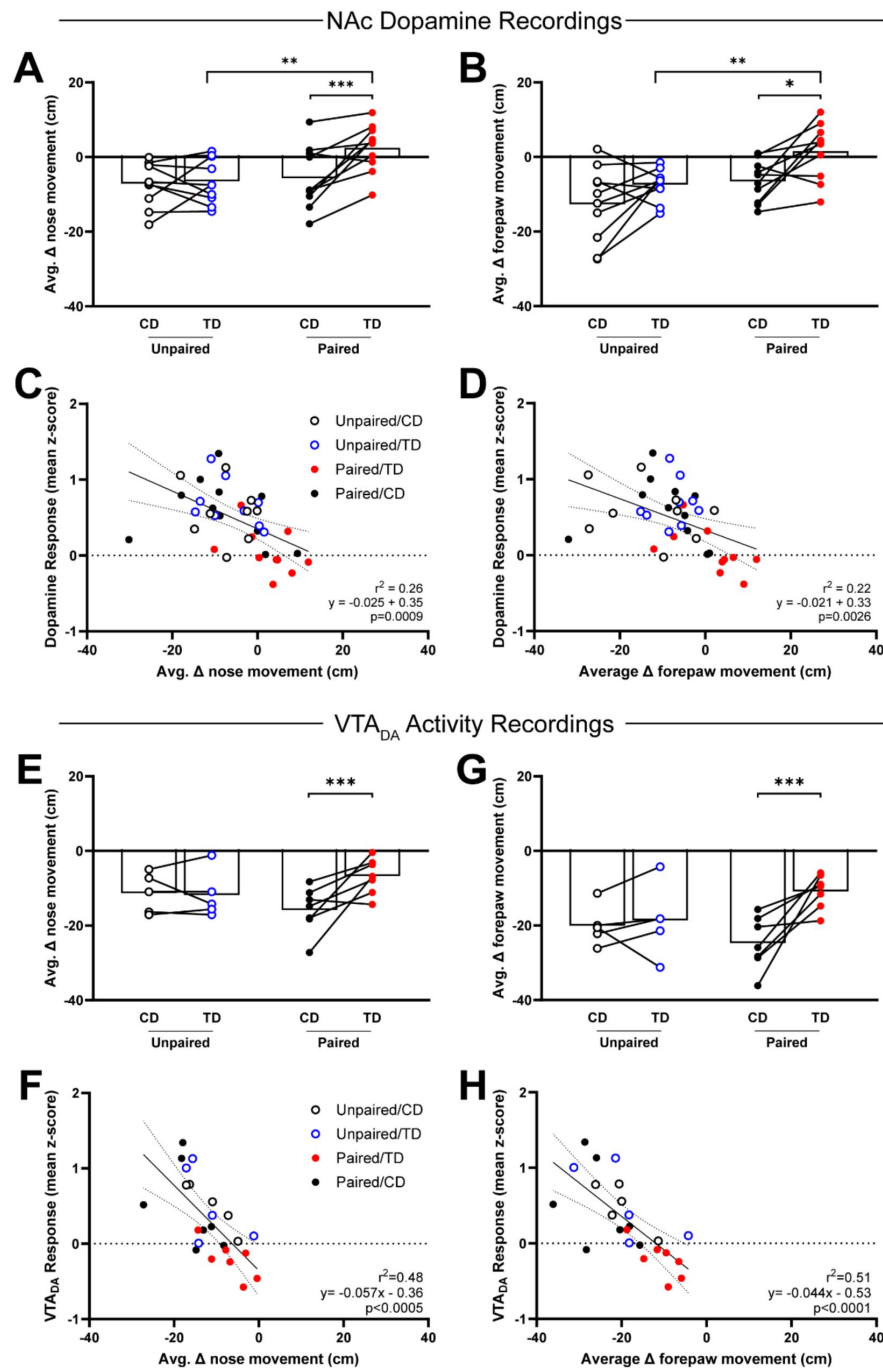


Figure 4.

Suppressed dopamine responses correlate with enhanced behavioral reactivity to intraoral sucrose delivery after CTA formation.

(A&B) Behavioral reactivity was quantified as the change in movement from baseline to infusion period. In Unpaired subjects, the average behavioral reactivity of nose (left) or forepaw (right) movement did not change from CD to TD. In Paired subjects, behavioral reactivity increased from CD to TD. (C) Relationship between behavioral reactivity of nose movement and mean z-score of NAC dopamine during infusion averaged by session. (D) Relationship between average change in behavioral reactivity of forepaw movement and mean z-score of VTA_{DA} activity responses during infusion averaged by session. Data in A, B, E & G are represented as means; * $p < 0.05$, ** $p < 0.01$; *** $p < 0.005$, Two-Way RM ANOVA with Uncorrected Fisher's LSD post-hoc. Line in C, D, F & H denote the linear relationship between parameters with dotted lines as 95% confidence intervals. P-value of linear regressions indicate slope's deviation from zero.

subsequent injection (**Fig. 5A**). To examine voluntary sucrose preference or avoidance, rats were given access to sucrose and water for two hours following each intraoral sucrose session (**Fig. 5A**). In a within-subjects comparison, intraoral sucrose infusions produced comparable VTA_{DA} activity responses in Unpaired rats across CD and all Extinction days (E1-E5, Unpaired: main effect of session: $F_{(2,60, 15.59)}=2.06$, $p>0.05$; **Fig. 5B** & 5C). In contrast, VTA_{DA} responses to intraoral sucrose were influenced by CTA formation and extinction (Paired: main effect of Day: $F_{(2.18, 15.27)}=10.32$, $p>0.005$; **Fig. 5B** & 5C). CTA formation suppressed sucrose-driven VTA_{DA} responses (Paired: C1 vs. E1, $p<0.005$; **Fig. 5B** & 5C) but VTA_{DA} responses were recovered to CD levels during with re-exposure (Paired: C1 vs. E2, $p>0.05$; **Fig. 5B** & 5C). To assess between subject differences across Days, we used an ROC to compare the distribution of VTA_{DA} responses on individual trials between Treatments (Unpaired vs Paired). VTA_{DA} responses to intraoral sucrose were discriminable on E1 ($AUC_{ROC} = 0.76$) and E2 ($AUC_{ROC} = 0.77$; **Fig. 5E-F**) but not during E3-E5 ($AUC_{ROC} \leq 0.63$).

Preference scores also reflected changing responses to sucrose. Paired rats showed an avoidance of sucrose relative to Unpaired rats (main effect of group $F_{(1,13)}=35.24$, $p<0.0001$; **Fig. 6A**) and sucrose preference scores of Paired rats changed across Extinction days (Treatment X Day: $F_{(4,52)}=33.14$, $p<0.0001$; **Fig. 6A**). Relative to Unpaired rats, Paired subjects had a significantly lower preference for sucrose on E1 ($p<0.0001$; **Fig. 6A**) and E2 ($p<0.0001$; **Fig. 6A**), which recovered to Unpaired levels by E3 ($p>0.05$; **Fig. 6A**). Importantly, sucrose preference was predicted by the earlier recording of VTA_{DA} responses to intraoral sucrose ($r^2=0.38$, slope=0.33, $p<0.0001$; **Fig. 6B**).

To confirm the recovery of VTA_{DA} responses after CTA extinction was due to presentations of intraoral sucrose without subsequent US exposure (malaise) and not simply the passage of time, we recorded VTA_{DA} activity from a separate group of rats using the Single-pairing CTA paradigm but with a delayed TD (**Supplementary Fig. 4**). Here, rats were re-exposed to intraoral sucrose seven days after CD (**Fig. 7A**); thus, the timing of TD was equated with E5 from the Single-pairing paradigm with Extinction (**Fig. 5A**). Intraoral sucrose evoked comparable VTA_{DA} responses in Unpaired rats between CD and TD (Unpaired: $t(4) = 0.97$, $p>0.05$; **Fig. 7B**). In contrast, dopamine responses were significantly suppressed in Paired rats on the delayed TD relative to CD (Paired: $t(4) = 10.24$, $p=0.0005$; **Fig. 7B**). Further, while individual VTA_{DA} responses to intraoral sucrose in Unpaired and Paired rats could not be discriminated on CD ($AUC_{ROC} = 0.59$), they were well discriminated on TD ($AUC_{ROC} = 0.84$, **Fig. 7C-F**).

Suppression of phasic dopamine responses in CTA scales with conditioning and extinction

To determine if phasic dopamine responses scale with the strength of a taste aversion, VTA_{DA} activity was monitored in a separate cohort of rats undergoing a Repeated-pairing CTA with Extinction paradigm (**Supplementary Fig. 5**). Here, three CD cycles with accompanying noncontingent injection and ‘off’ days (C1-C3) were administered to Unpaired and Paired rats followed immediately by Extinction days E1-E8. In addition, sucrose preference tests were conducted after each Extinction day (**Fig. 7A**). In Unpaired rats, average VTA_{DA} activity was comparable across all Conditioning (C1-C3) and Extinction days (E1-E8, Unpaired: main effect of day: $F_{(3,88, 23.30)}=1.99$, $p>0.05$; **Fig. 8B-D**). In contrast, Paired rats showed significant changes in VTA_{DA} responses to sucrose across the paradigm (Paired: main effect of day: $F_{(2.86, 19.98)}=16.07$, $p<0.0001$; **Fig. 8B-D**). In Paired rats, VTA_{DA} responses were suppressed on C3 through E4 (Paired: Relative to C1: C3-E4 $p<0.05$; **Fig. 8B-D**) but not subsequent Extinction days (Paired: Relative to C1: E5-E8 $p>0.05$; **Fig. 8B-D**). Between subject differences were also evident; responses to intraoral sucrose were discriminable between Unpaired and Paired groups on C3 ($AUC_{ROC} = 0.88$) and E1-E6 ($AUC_{ROC} >0.7$; see **Fig. 8E-G** for individual AUC_{ROCS}).

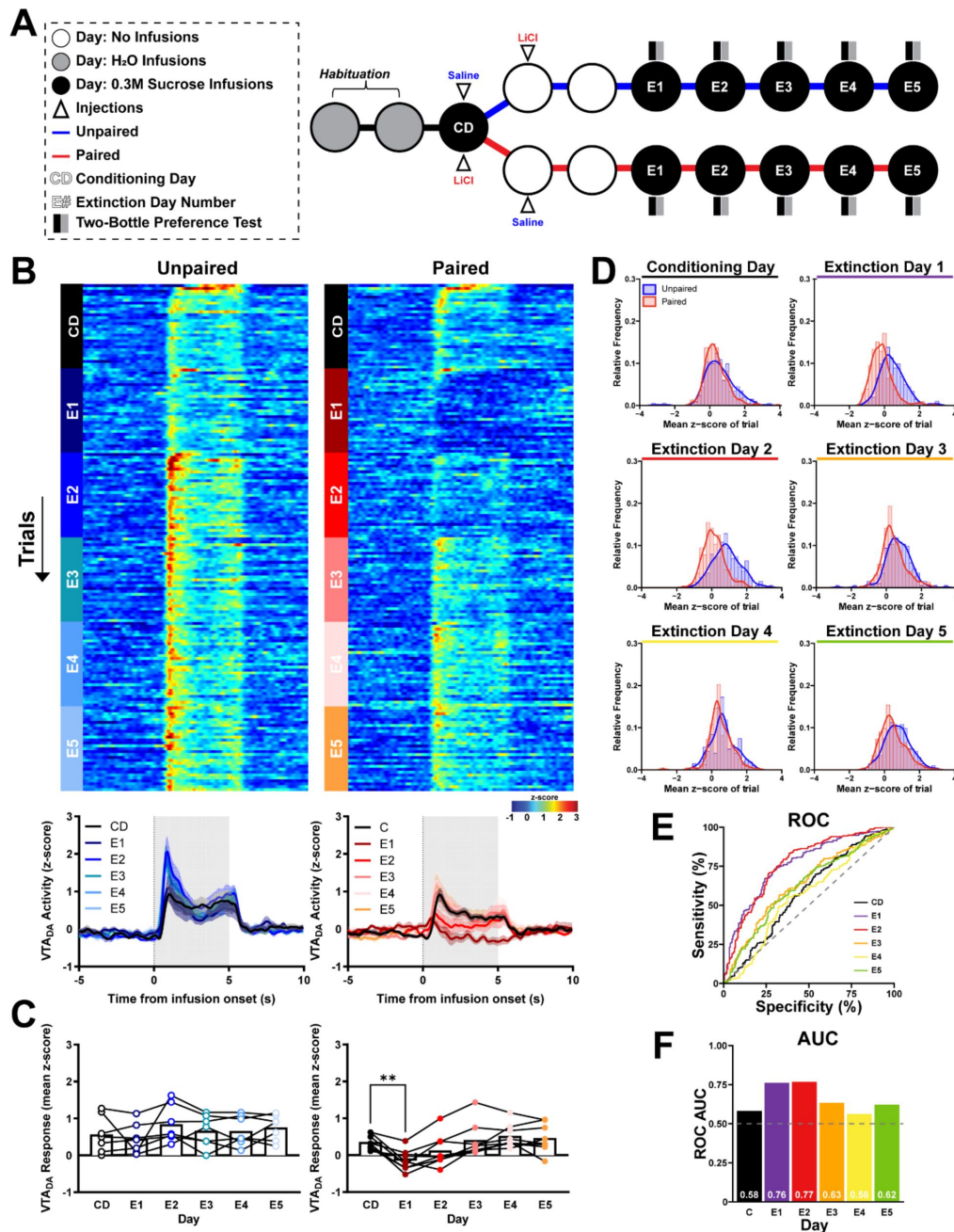


Figure 5.

Sucrose exposure under extinction conditions ameliorates CTA's suppression of phasic dopamine responses to intraoral sucrose.

(A) Schematic of the CTA paradigm consisting of a single-pairing of LiCl or saline pairing to sucrose followed by five consecutive sessions intraoral sucrose not subject to additional US (malaise) exposure. CTA training as in [Figure 3](#) was conducted followed by five Extinction sessions (E1-E5), in which rats received intraoral infusions (parameters identical to TD) without any additional injections. (B) VTA_{DA} across trials and sessions before and after intraoral sucrose delivery onset. *Top*: VTA_{DA} activity (in z-score) across 30 trials of the CD and E1-E5 sessions. *Bottom*: Average traces of VTA_{DA} activity aligned to the onset of intraoral delivery of sucrose. (C) Mean z-score during 5s intraoral delivery of sucrose. (D) Relative frequency histogram of VTA_{DA} responses to sucrose on all test days for every trial reported as mean z-score for both Unpaired and Paired subjects. (E) ROC of relative frequency distributions of mean z-score acquired from each trial of on all test sessions between treatment groups. (F) Plotted area under the curve (AUC_{ROC}) values of E. Data in C are represented as means; ** $p < 0.01$, One-Way RM ANOVA with Dunnett's multiple comparisons test post hoc.

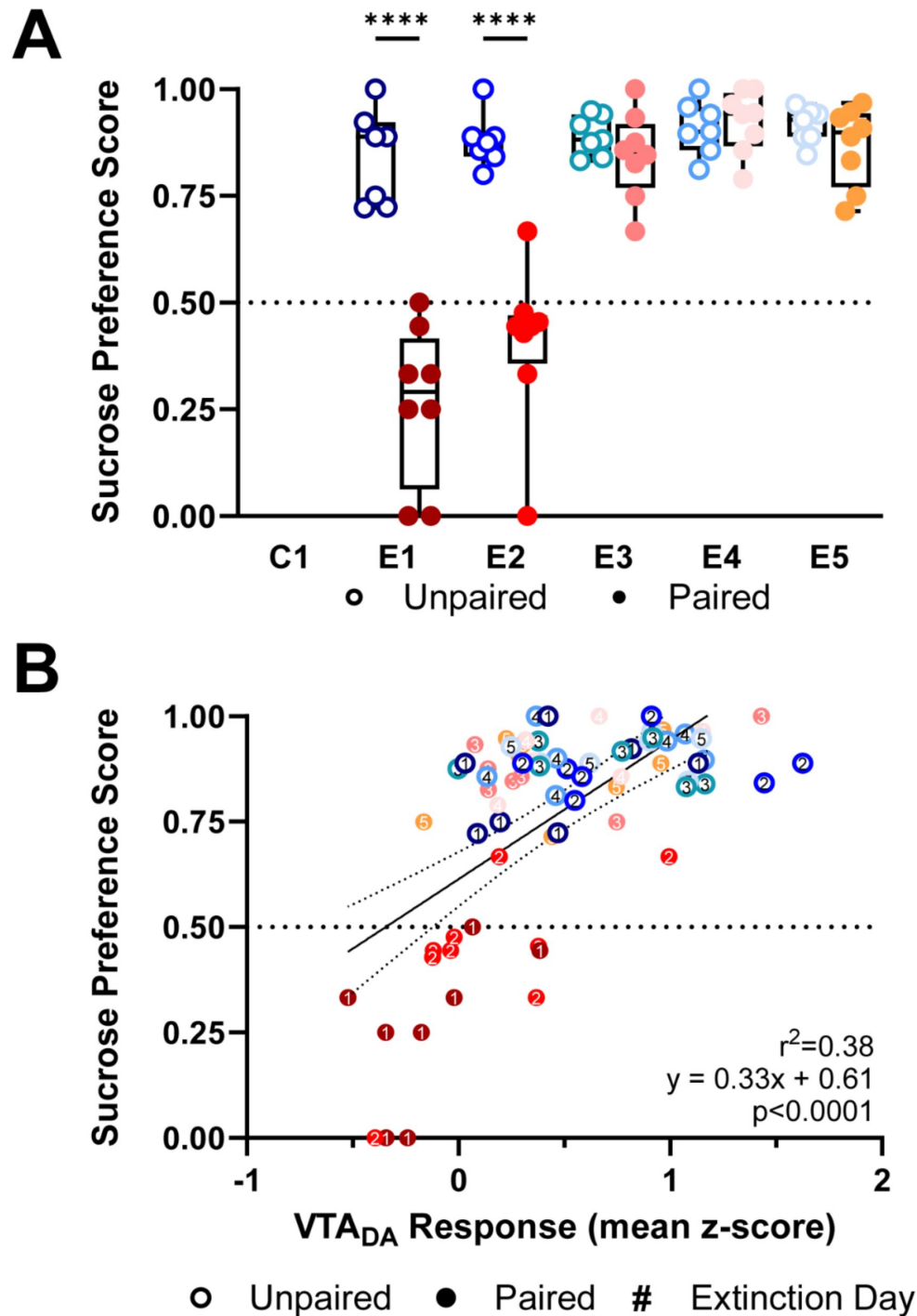


Figure 6.

Suppressed dopamine responses to sucrose predict conditioned taste avoidance.

(A) Each Extinction session was followed by a two-hour Two-Bottle Preference test with access to both sucrose and water. Sucrose preference scores were calculated as the percent of sucrose solution consumed from the sum of sucrose and water consumed. Paired (red, closed circles) rats showed decreased sucrose preference on E1 and E2 relative to Unpaired rats (blue, open circles). (B) Average VTA_{DA} responses during intraoral sucrose delivery were positively correlated to sucrose preference scores calculated from the Two-Bottle Preference test. Data in A are represented as means; **** $p < 0.001$, Two-Way RM ANOVA with Šidák multiple comparisons test post hoc. Line in B denotes the linear relationship between parameters with dotted lines as 95% confidence intervals. P-value of linear regression indicates slope's deviation from zero.

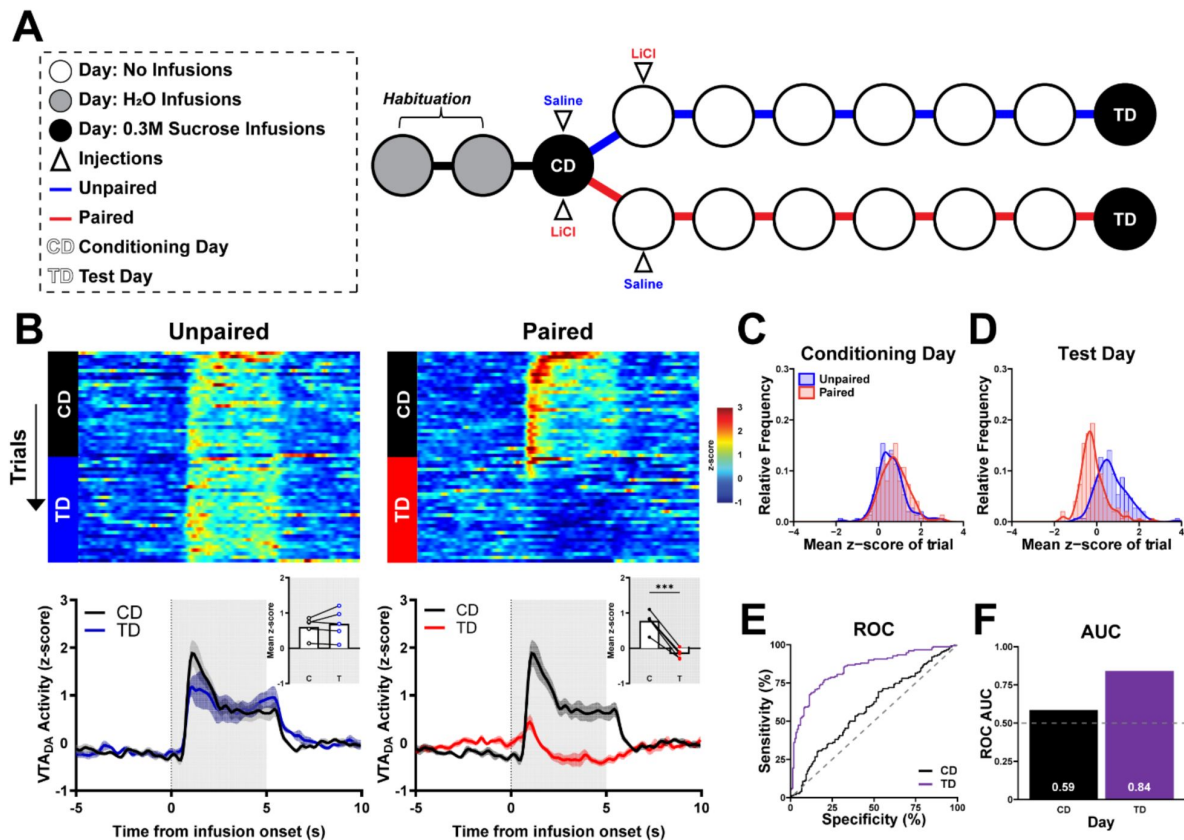


Figure 7.

Delayed testing after sucrose pairing to LiCl-induced malaise suppresses dopamine response to sucrose.

(A) Schematic of the Delayed-test CTA paradigm. CTA training as in [Figure 3](#) was conducted. To match the timeline of Single-pairing CTA with Extinction, rats had five additional untreated days followed by Test Day (TD), which corresponded to the delay from CD to E5. On TD, rats received intraoral infusions (parameters identical to E1). (B) VTA_{DA} across trials and sessions before and after intraoral sucrose delivery onset. *Top*: VTA_{DA} activity (in z-score) across 30 trials during CD and TD sessions. *Bottom*: Average traces of VTA_{DA} activity aligned to the onset of intraoral delivery of sucrose. *Inset*: Average z-score quantified during 5s intraoral delivery of sucrose. (C-D) Relative frequency histogram of dopamine release responses to sucrose on CD and TD for every trial reported as mean z-score for both Unpaired and Paired subjects. (E) ROC of relative frequency distributions of mean z-score acquired from each trial of on CD and TD between treatment groups. (F) Plotted area under the curve (AUC_{ROC}) values of E. Data in B are represented as means; *** $p < 0.005$, paired t-test.

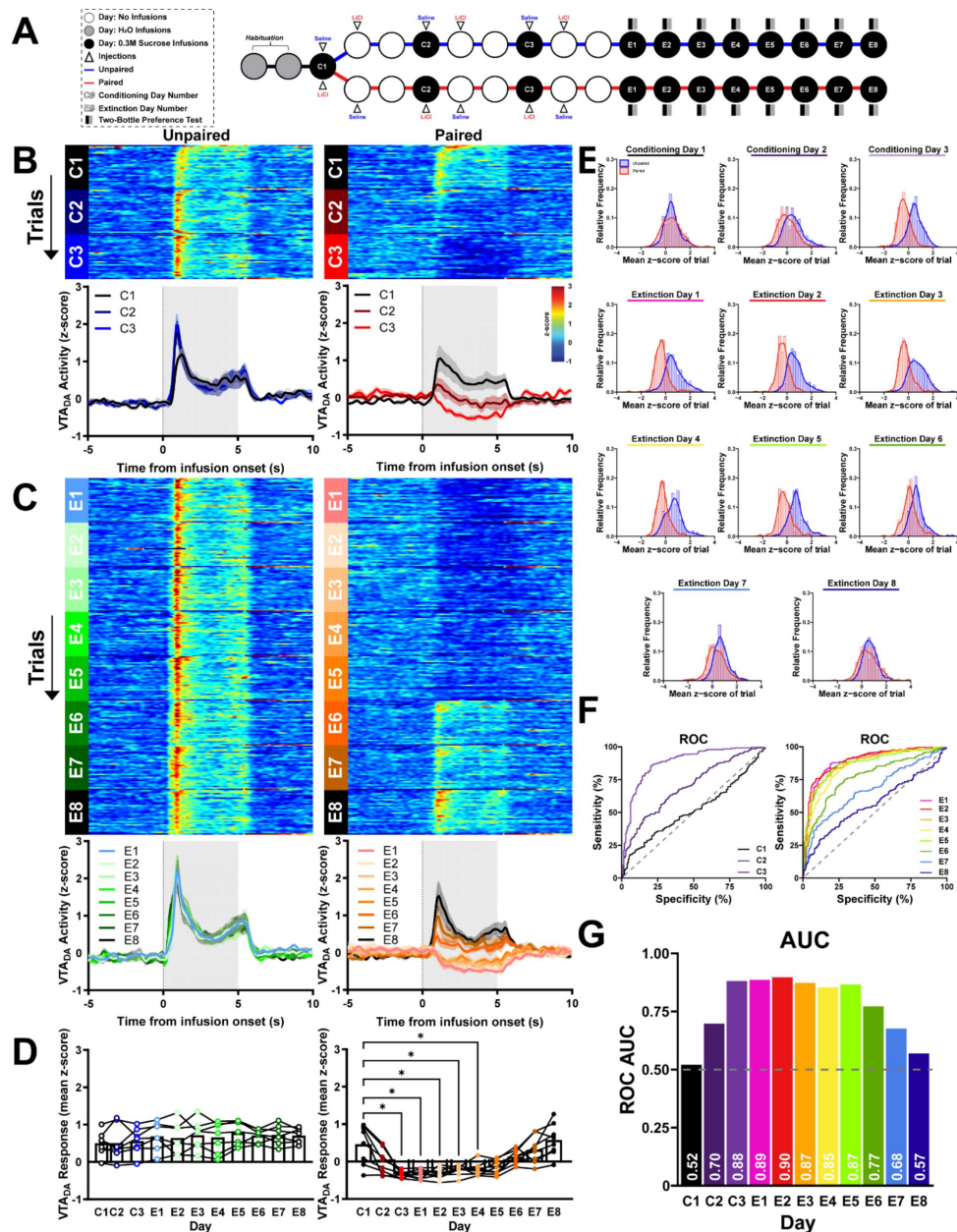


Figure 8.

Phasic dopamine responses to intraoral sucrose scale to the strength of the CTA.

(A) Schematic of the Repeated-pairing CTA paradigm. The three-day CTA conditioning process (as in Figure 3) was repeated three times (C1-C3). Following conditioning, subjects received eight daily Extinction sessions (E1-E8) with intraoral infusions of sucrose followed by no US (malaise) exposure. After the behavioral sessions, rats had access to sucrose and water consumption for two-hours (Two-Bottle Preference Test). (B) VTA_{DA} across trials and sessions before and after intraoral sucrose delivery onset. *Top*: Heat maps for VTA_{DA} activity (in z-score) for 30 trials in each session during the three conditioning sessions (C1-C3). *Bottom*: Associated average traces of VTA_{DA} activity aligned to the onset of intraoral delivery of sucrose. (C) Conventions as in B but for all eight non-reinforced sucrose sessions (E1-E8). (D) Average z-score quantified during 5s intraoral delivery of sucrose for all conditioning and subsequent test sessions. (E) Relative frequency histogram of VTA_{DA} responses to sucrose on all test days for every trial reported as mean z-score for both Unpaired and Paired subjects. (F) ROC of relative frequency distributions of mean z-score acquired from each trial of on all conditioning (left) and extinction (right) days between treatment groups. (G) Plotted area under the curve (AUC_{ROC}) values of F. Data in D are represented as means; * $p < 0.05$, One-Way RM ANOVA with Dunnett's multiple comparisons test post hoc.

Behavioral reactivity was simultaneously recorded with VTA_{DA} activity. Intraoral sucrose delivery produced comparable behavioral responses in Unpaired rats across all days (Unpaired: main effect of day: $F_{(3,19, 19,12)}=0.74$, $p>0.05$; **Fig. 9A**). In contrast, Paired subjects showed shifts in behavioral responses across conditioning and extinction days (Paired: main effect of day: $F_{(4,02, 28,15)}=7.70$, $p=0.0003$; **Fig. 9A**). Relative to the first conditioning day, head movement was elevated from C3 through E3 and decreased with subsequent extinction sessions (Paired: Relative to C1: C3, E2, E3 $p<0.05$, E1 $p<0.005$; **Fig. 9A**). VTA_{DA} activity and behavioral reactivity were negatively correlated across all rats and days ($r^2=0.43$, slope=-9.52, $p<0.0001$; **Fig. 9B**).

After each extinction session, we conducted a sucrose preference test. Paired rats avoided sucrose compared to Unpaired rats (main effect of group: $F_{(1,13)}=59.78$, $p<0.0001$; **Fig. 9C**). Specifically, Paired rats had a lower sucrose preference on E1 through E5 (E1&E2 $p<0.0001$, E3 $p<0.005$, E4 $p=0.0001$, E5 $p<0.05$; **Fig. 9C**). Similar to behavioral reactivity but with more predictive power, VTA_{DA} responses were tightly correlated with post-session sucrose preference ($r^2=0.59$, slope=0.48, $p<0.0001$; **Fig. 9D**).

Discussion

Unexpected rewards and reward-predictive cues evoke phasic dopamine activity and NAc dopamine release (Schultz, 1998). Responses to aversion remain more controversial (McCutcheon et al., 2012; Morales and Margolis, 2017) and may differ based on sensory transduction pathways (McCutcheon et al., 2012) and whether the aversive outcome can be avoided (Goedhoop et al., 2022; Oleson et al., 2012) – which contribute to competing hypotheses that phasic dopamine responses signal either valence (reward versus aversion) or salience (reward and aversion (Kutlu et al., 2021)). Here, we circumvent common obstacles to determining responses to aversive stimuli by capturing real-time dopamine responses to the same taste stimulus – sucrose – and modulating its valence through conditioning. Intraoral delivery of sucrose eliminated avoidance as a potential confound. Measurement of behavior ensured that rats were responding to the intraoral infusion, eliminating differences in stimulus salience as a contributing factor. A key aspect of our paradigms is that all rats had identical exposure to sucrose and illness with the temporal proximity of sucrose and illness being the lone difference. Ultimately, we show that dopamine responses to sucrose scale with valence across acquisition and extinction of a CTA.

Phasic dopamine activity differentially encodes the valence of primary appetitive and aversive taste stimuli

Dopamine responses to primary aversive stimuli are mixed. Noxious stimuli have been reported to increase the firing rate of dopamine cell bodies in the VTA (Anstrom et al., 2009; Anstrom and Woodward, 2005) and dopamine release in the NAc (Budygin et al., 2012; Kiyatkin, 1995; Mikhailova et al., 2019; Young, 2004; Young et al., 1993). Conversely, others have reported that noxious stimuli inhibit VTA_{DA} firing (Brischoux et al., 2009; Ungless et al., 2004) and NAc dopamine release (Mantz et al., 1989; Wenzel et al., 2015) or that responses vary depending on the VTA_{DA} subpopulation or dopamine release site (Brischoux et al., 2009; de Jong et al., 2019; Zweifel et al., 2011). Aversive stimuli can drive excitatory inputs to the VTA (Amo et al., 2024; Faget et al., 2024). With respect to taste stimuli specifically, some reports indicate decreased (Roitman et al. 2008; Twining et al., 2015), while others report increased (Bassareo et al., 2002; Kutlu et al., 2021) dopamine responses to bitter quinine. Here, we used intraoral delivery of appetitive and aversive taste stimuli to circumvent confounds including active or passive avoidance, and food or water deprivation. Our findings are similar to prior work from our group (Hsu et al., 2020; McCutcheon et al., 2012; Roitman et al., 2008): intraoral sucrose increased NAc dopamine release while the response in the same rats to quinine was significantly lower. Neuroanatomical data support projections to the VTA from relatively early nodes of the

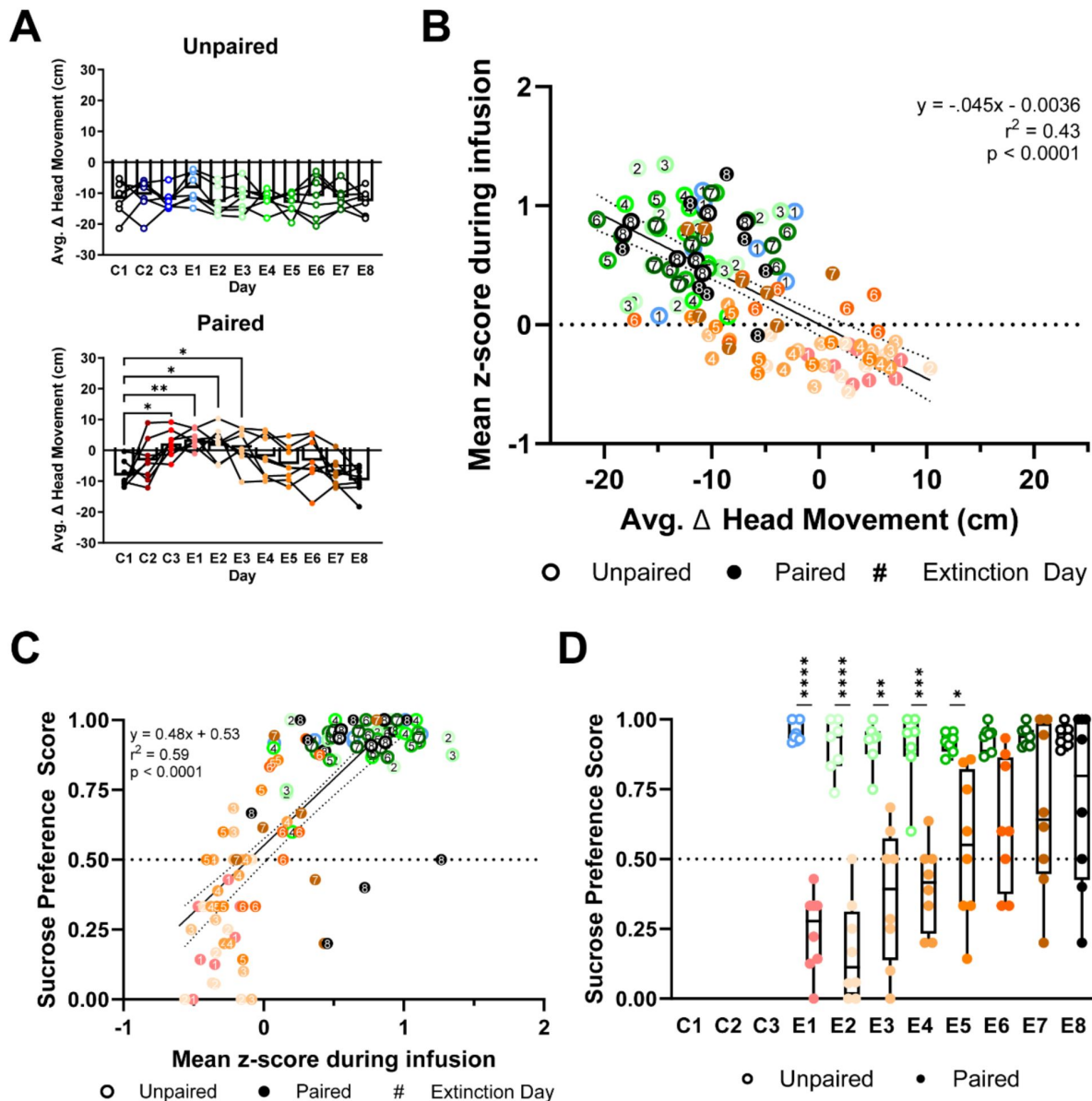


Figure 9.

Intraoral sucrose driven dopamine responses are negatively correlated to behavioral reactivity and predict sucrose preference.

(A) Head movement to intraoral sucrose averaged across trials. *Top*: Unpaired rats showed comparable average change in head movement behavioral reactivity across testing sessions. *Bottom*: Paired subjects showed an increase in average head movement responses to intraoral sucrose from C1 to C2-E3. (B) Relationship between average head movement behavioral reactivity and VTA_{DA} responses (in mean z-score) during infusion averaged by session. (C) After each E1-E8 session, rats were administered the Two-Bottle Preference test. Paired (red, closed circles) rats showed decreased sucrose preference on E1-E5 relative to Unpaired rats (blue, open circles). (D) Average VTA_{DA} responses during intraoral sucrose delivery were positively correlated to sucrose preference scores calculated from the Two-Bottle Preference test. Data in A are represented as means; * $p < 0.05$, ** $p < 0.01$, One-Way RM ANOVA with Dunnett's multiple comparisons test post hoc. Data in C are represented as means; * $p < 0.05$, ** $p < 0.01$, *** $p < 0.005$, **** $p < 0.001$, Two-Way RM ANOVA with Šidák multiple comparisons test post hoc. Line in B & D denotes the linear relationship between parameters with dotted lines as 95% confidence intervals. P-value of linear regression indicates slope's deviation from zero.

gustatory pathway (e.g. parabrachial nucleus; PBN) and specifically sucrose and quinine responsive neurons in the PBN (Boughter et al., 2019 [DOI](#)). It remains to be determined if sucrose- and quinine- responsive PBN neurons synapse onto different VTA cell types (e.g. dopamine neurons for the former and GABA neurons for the latter).

Differences in stimulus salience could contribute to the dopamine responses observed here. Indeed, recent work has supported dopamine alignment with the encoding of stimulus salience (Chen and Bruchas, 2021 [DOI](#); Kutlu et al., 2021 [DOI](#)), to contribute to prioritizing attentional processes towards stimuli of high intensity. Further, human VTA activation is greater for novel, more highly salient stimuli than to familiar stimuli (Bunzeck and D  zel, 2006 [DOI](#)). Here, behavioral responses to quinine were greater than those for sucrose supporting quinine as a highly salient stimulus. Nonetheless, the strong inverse relationship between behavioral reactivity and dopamine release observed here, and particularly the lack of a dopamine response to quinine refutes the idea that phasic dopamine encodes the salience of the stimulus.

Phasic dopamine activity differentially encodes appetitive versus aversive sucrose

The differential dopamine response to appetitive and aversive tastes supports affective encoding by the mesolimbic system. However, sucrose and quinine act on different taste receptors (Schier and Spector, 2019 [DOI](#)) and may access the VTA via parallel pathways (Boughter et al., 2019 [DOI](#)). We therefore employed a CTA paradigm to measure dopamine responses to the same stimulus when the stimulus was appetitive versus aversive. While dopamine responses were identical across groups on CD (when rats were initially na  ve to intraoral sucrose), pairing intraoral sucrose infusions with LiCl injection caused changed, suppressed dopamine responses to intraoral sucrose when it was next administered (TD). Importantly, the temporal relationship between illness and initial sucrose exposure, as in seminal studies (Smith and Roll, 1967 [DOI](#)), determined both the formation of a CTA (as evidence in behavioral reactivity measures) and suppressed dopamine activity and release. Indeed, LiCl injection administered 24hr after intraoral sucrose (Unpaired) failed to modulate dopamine signaling. Recent work has suggested that dopamine release from terminals within the NAc could result from intra-NAc processes that are independent of cell body activity (Cachope and Cheer, 2014 [DOI](#); Mohebi et al., 2023 [DOI](#); Threlfell et al., 2012 [DOI](#)). However, we found similar modulation of dopamine responses in Paired rats when recordings captured NAc dopamine release or VTA_{DA} cell body activity. Our data are more consistent with the good agreement between dopamine cell body activity, release, and post-synaptic signaling reported by others (Lee et al., 2020 [DOI](#)). Local GABA neurons innervate and suppress VTA_{DA} neurons through neurotransmission at GABA_A receptors, which can pause VTA_{DA} activity and downstream dopamine release in the NAc (Mathon et al., 2005 [DOI](#); Tan et al., 2012 [DOI](#); Van Zessen et al., 2012 [DOI](#)). Understanding local GABA dynamics will be an intriguing direction to identify mechanisms for suppression of dopamine signaling to innately aversive tastes as well as those that have acquired aversion through conditioning.

Throughout our studies, we found an inverse relationship between dopamine release/activity and behavioral reactivity to intraoral infusions. Decreases in dopamine activity can lead to an uncoupling of dopamine and high affinity dopamine D2-like receptors expressed on medium spiny neurons (Dreyer et al., 2010 [DOI](#)). In turn, this could contribute to the different patterns of NAc activity reported for rewarding versus aversive primary (Roitman et al., 2005 [DOI](#)) and conditioned tastes (Roitman et al., 2010 [DOI](#); Wheeler et al., 2008 [DOI](#)). As manipulations of NAc activity can influence taste reactivity (Morales and Berridge, 2020 [DOI](#)), differential dopamine responses to sucrose in rats with versus without CTA likely applies a critical filter for NAc processing in the service of oromotor output.

In our purely classical conditioning CTA paradigm, taste (i.e., intraoral sucrose) served as a conditioned stimulus for visceral malaise. Other aversive conditioning paradigms have shown development of decreased dopamine responses to cues predictive of primary aversive stimuli (Goedhoop et al., 2022; Zhuo et al., 2024). There are unique aspects of CTA worth emphasizing. As we've shown here, CTA can form with just one taste-illness pairing. In addition, CTA tolerates relatively long delays between the conditioned (taste) and unconditioned stimulus (LiCl). While LiCl was injected immediately after the conditioning session, this interval exceeded 30 minutes from the initial taste exposure. Moreover, indices of the malaise induced by LiCl do not emerge for many minutes, e.g. ~5min for lying on belly (Aguilar-Rivera et al., 2020), ~20min for pica (Aguilar-Rivera et al., 2020), and ~20min for decrease in core body temperature (Bernstein et al., 1992; Guimaraes et al., 2015). Thus, taste and illness memory traces must be integrated over many minutes. The PBN is essential for CTA formation (Spector et al., 1992) and contains neurons that process gustatory and illness information (Carter et al., 2015). The amygdala is also critical for CTA acquisition (Gao et al., 2023; Inui et al., 2019; Morin et al., 2021; Morris et al., 1999; St. Andre and Reilly, 2007). The same neurons within the basolateral nucleus of the amygdala are responsive to both novel taste and LiCl (Barot et al., 2008; Zimmerman et al., 2024). The rostral tegmental nucleus (RMTg) is a robust source of GABAergic input onto VTA_{DA} neurons and suppresses their firing (Balcita-Pedicino et al., 2011; Bourdy et al., 2014). Primary aversive stimuli, including LiCl, increase RMTg neuronal firing (Li et al., 2019). Further, re-exposure to LiCl-paired saccharin increases cFos expression in the RMTg (Glover et al., 2016). Likewise, LiCl injections suppress electrically-evoked NAc dopamine release (Fortin et al. 2016). How taste and illness traces and their integration ultimately modulate the mesolimbic system is an important ongoing direction.

Suppression of dopamine signaling scales with CTA expression and extinction

Multiple conditioning trials caused greater suppression in dopamine activity evoked by the sucrose CS. This has been previously reported for audiovisual cues that predict discrete, noxious stimuli (e.g. air puff (Zhuo et al., 2024) and foot shock (Wilkinson et al., 1998)). Here, greater suppression occurred with each taste-illness pairing over just three conditioning trials spaced days apart. Taste, relative to other sensory stimuli, serves as a uniquely strong predictor for illness (Garcia and Koelling, 1966). After establishing CTA, repeated re-exposure to the conditioned stimulus without subsequent malaise drives CTA extinction (Cantora et al., 2006; Mickley et al., 2004; Nolan et al., 1997), allowing for new, contextual knowledge to conflict with the original association between the taste and negative post-ingestive outcome (Bouton and Bolles, 1979). We found that the suppressed dopamine response to sucrose gradually returned to conditioning day levels after repeated extinction sessions. The number of extinction sessions needed for the restoration of dopamine signaling scaled with the number of conditioning sessions (i.e., two extinction sessions for one pairing and five extinction sessions for three pairings). The return of dopamine responses to conditioning day levels was not simply due to the passage of time but rather required CS re-exposure. As with all experiments herein, dopamine responses were negatively correlated with behavioral reactivity across acquisition and extinction. Importantly, dopamine responses to intraoral sucrose were also predictive of home cage sucrose preference across extinction. The infralimbic (IL) cortex plays a role in extinction learning (Quirk et al., 2006). Stimulation of the IL suppresses aversive taste reactivity to sucrose after it had been paired with LiCl (Hurley and Carelli, 2020). Thus, while initial taste-LiCl associations may be due to subcortical (e.g. PBN, RMTg, amygdala) influences on sucrose-evoked VTA_{DA} activity and NAc dopamine release, extinction of the suppressed dopamine response to sucrose may well engage top-down cortical processes.

Our studies support a role for dopamine in differentially encoding taste valence. It is important to recognize that there is considerable heterogeneity among dopamine neurons (Phillips et al., 2022). Indeed, VTA neurons that release both glutamate and dopamine are important for

aversion, and it is the dopamine signal from these neurons that is critical for this process (Warlow et al., 2024 [↗](#)). The technical approaches used here to measure VTA_{DA} activity and dopamine release in the NAc average responses across an unknown number of neurons. Thus, while our results suggest that most dopamine neurons decrease their activity and release of dopamine in the NAc to primary and conditioned aversive taste, subpopulations likely play different and critical roles in appetitive and aversive responses. Combining molecular markers and defining input/output relationships for VTA dopamine neurons will be critical for further advancing an understanding of dopamine and aversion. A range of clinically used drugs – from anti-obesity treatments to chemotherapeutics – induce malaise and can lead to the development of CTAs and a reduction in motivation. Thus, further understanding of how these associative processes influence dopamine signaling is important to yield therapeutic approaches with greater efficacy and improved adherence.

Materials & methods

Subjects

Male and female (intact, naturally cycling) Long Evans rats (>250g) were bred from females heterozygous for expressing Cre recombinase under the control of the tyrosine hydroxylase promoter (TH:Cre) (Rat Research Resource Center, RRRC No. 659; Witten et al., 2011 [↗](#)) and male wildtype rats (Charles River Laboratories). Offspring were genotyped for Cre (Transnetyx, Inc.) via tissue from ear punches. TH:Cre+ (n=31 males, 22 females) rats were used for experiments where VTA_{DA} activity was measured. TH:Cre-(n=13 males, 14 females) rats were used for experiments where NAc dopamine release was measured. Rats were individually-housed in a temperature- and humidity-controlled room and on a 12:12h light:dark schedule (lights on 0700h). All experiments were conducted during the light cycle. Food and water were provided *ad libitum*. All studies were conducted in accordance with the National Institutes for Health Guide for the Care and Use of Laboratory Animals and approved by the Animal Care Committee at the University of Illinois Chicago.

Viruses

Adeno-associated viruses (AAVs) packaged with fluorescent sensors were used for in vivo fiber photometry. AAV1.hSyn.Flex.GCaMP6f.WPRE.SV40 (5 x 10¹²GC/mL, Addgene) was used to detect calcium (Ca²⁺; Chen et al., 2013 [↗](#)). AAV9.hSyn.GRAB_DA2h (1 x 10¹³GC/mL, Addgene) was used to detect dopamine (Sun et al., 2020 [↗](#)).

Surgeries

Rats were anesthetized with isoflurane, injected with an analgesic (1mg/kg meloxicam, subcutaneous), and placed in a stereotaxic instrument. To record VTA_{DA} activity, 1μL of AAV1.hSyn.Flex.GCaMP6f.WPRE.SV40 was injected into the VTA (Anterior/Posterior (AP): - 5.40mm, Medial/Lateral (ML): +0.70mm, Dorsal/Ventral (DV): -8.15mm, relative to Bregma) of TH:Cre+ rats at a rate of 0.1μL/min. A 5-min post-injection period to allow for diffusion followed. A fiber optic (flat 400-μm core, 0.48 numerical aperture, Doric Lenses Inc.) was then implanted just dorsal (DV: -8.00mm) to the virus injection. To measure dopamine release, 1μL of AAV9-hsyn-GRAB_DA2h was injected into the NAc lateral shell (AP: +1.5mm, ML: +2.5mm, DV: -8.0mm relative to Bregma) of TH:Cre-rats. A fiber optic was implanted just dorsal to the virus injection (DV: -7.9mm). For all surgeries, an intraoral catheter was also implanted. Catheters, made from ~6cm length of PE6 tubing (Scientific Commodities, Inc.) threaded through a Teflon washer, were inserted via hypodermic needle just lateral to the first maxillary molar. The needle was guided subcutaneously and exteriorized out of the incision at the top of the head. Implants were cemented to skull screws using Metabond (Parkell, Inc.) and dental acrylic. To allow for recovery and construct expression, experiments began at minimum 21-days after surgery (Fig. 1A [↗](#)).

Fiber photometry recording

Light-emitting diodes (LEDs; Doric Lenses) emitted 465nm (Ca^{2+} or dopamine-dependent) and 405nm (Ca^{2+} -independent) wavelengths. At 30 μW emission power, frequencies of 465nm and 405nm light were sinusoidally modulated at 210Hz and 330Hz, respectively. Light was coupled to a filter cube (FMC4, Doric Lenses) and converged into an optical fiber patch cord mated to the fiber optic implant of the rat. Fluorescence was collected by the same fiber/patch cord and focused onto a photoreceiver (Visible Femtowatt Photoreceiver Model 2151, Newport). A lock-in amplifier and data acquisition system (RZ10X; Tucker Davis Technologies), was used to demodulate the fluorescence due to 465nm and 405nm excitation. Transistor-transistor logic (TTL) signals from the system operating behavioral events (Med-Associates, Inc.) were sent to the data acquisition system, timestamped, and recorded along with fluorescence using software (Synapse Suite, Tucker Davis Technologies).

After acquisition, data were Fourier-transformed and the 405nm excitation signal was subtracted from the 465nm excitation signal to account for movement artifacts and photobleaching ($\Delta F/F$). The subtracted signal was smoothed using a custom fifth order bandpass Butterworth filter (cutoff frequencies: 0.05Hz, 2.25Hz) and returned to the time domain using custom scripts written in MATLAB (version R2022b, Mathworks; code available on Github). To compare changes in fluorescence across recording sessions and across rats in different experimental groups, the processed signal for each recording session was normalized to the session's average fluorescence and converted to z-scores. The normalized signal was then aligned to events of interest (**Fig. 1B** [↗](#)).

Intraoral Infusion Protocols

Before each session, intraoral catheters were flushed with distilled water to ensure patency. All training and experimental sessions took place in standard chambers (ENV-009A-CT, Med Associates Inc.) or custom-made cylindrical chambers for behavioral reactivity recordings. At the start of a session, a fluid line from a syringe containing a test solution was gravity-fed into a 2-way solenoid valve (The Lee Company) and the fluid line exiting the solenoid was connected to the intraoral catheter. A TTL from Med Associates software and hardware triggered the opening and closing of the solenoid valve. Rats were habituated to receive 30 brief (5s, 200 μL) intraoral infusions of water repeated at varying intertrial intervals (35-55s) for two days. Intraoral delivery ensured that rats tasted the stimulus but were not obligated to consume it. This session structure was used throughout all experimental conditions detailed below.

Primary taste

Rats received intraoral infusions of either 0.3M sucrose or 0.001M quinine and received the other solution the following day with solution order counterbalanced across rats (**Fig. 1E** [↗](#)).

Single-pairing CTA

After habituation, all rats underwent a three-day training cycle. The first day was Conditioning Day (CD), where rats received intraoral infusions of 0.3M sucrose and were randomly assigned to either the Paired or Unpaired group. Immediately after the CD session, rats were injected intraperitoneally (i.p.; 20mL/kg) with either 0.15M LiCl for Paired rats or 0.15M NaCl for Unpaired rats. On the second day, rats did not receive intraoral infusions. Instead, they were injected with the counterbalanced treatment and returned to the home cage. On the third day, rats were left undisturbed in their home cage. This three-day CTA training cycle was followed by the Test Day (TD), in which rats received a session of intraoral infusions of sucrose with no subsequent injections.

Single-pairing CTA with Extinction

Treatment of Paired and Unpaired animals was the same as Single-pairing CTA above except that TD served as the first (E1) of five extinction sessions (E1-E5). For each extinction day, a session of intraoral infusions was administered as on TD. Extinction sessions were administered on consecutive days.

Delayed-test CTA

The same procedure as the Single-pairing CTA paradigm was used. However, TD was not performed until seven days after CD, to match the interval from CD to E5 from the Single-pairing CTA with Extinction condition.

Repeated-pairing CTA with Extinction

Paired and Unpaired animals underwent three conditioning cycles (C1-C3) of the procedure described in Single-pairing CTA. Following the conclusion of the third cycle, extinction sessions were administered for 8 consecutive days (E1-E8).

Two-Bottle Preference Test

After each Extinction session, rats were returned to the home cage and given 2-hour access to two sipper bottles, one containing 0.3M sucrose and the other water. Bottle locations were switched after 1-hour to avoid a side bias. Change in bottle weight from before to after 2-hour access was used to calculate sucrose preference as sucrose consumed (g) / total fluid consumed (g).

Behavioral Recordings

Behavior in all sessions was video captured at 10 fps for later analysis. In the Primary taste and Single-pairing CTA conditions, rats were recorded in a cylindrical chamber with a clear floor using an adjacent video camera (Teledyne FLIR, Model: BFS-U3-13Y3M-C) that captured behavioral reactivity via the reflection from an angled mirror positioned under the chamber (below view, Grill and Norgren, 1978a [↗](#)). In the studies of Single- and Repeated-pairing CTA with Extinction and Delayed-test CTA, rats were recorded in standard Med Associates chambers with a camera (Wo-We Webcam 720P) positioned above the chamber (above view). Video was monitored using recording software (below view: Teledyne FLIR, Spinnaker SDK; above view: Synapse, Tucker Davis Technologies) and stored for off-line analysis. A custom Med Associates program, linked to the camera's operation, sent TTLs to time stamp the onset and offset of the video recordings on the fiber photometry software (Synapse, Tucker Davis Technologies) to later sync video to photometry recordings.

Behavioral reactivity was calculated as the change in movement from the 5s pre-infusion period to the 5s intraoral infusion. Positional coordinates for body parts of the rat were obtained using the open-source deep-neural network toolbox, DeepLabCut ([Mathis et al., 2018 ↗](#)). A custom DeepLabCut model was developed and implemented to track the nose, forepaws, and chamber legs (for scale) in the below view recordings. For above view recordings, a different DeepLabCut model was developed to track the ears and chamber corners. Custom MATLAB codes were employed to calculate the midpoint between the ears (reported as head position) and track distance moved during specified time points, using the coordinates from stationary landmarks (i.e., chamber legs and corners) to normalize distances from pixels to metric units. Any coordinates with model likelihood estimates below 95% were excluded ($\leq 9\%$ of timepoints) and replaced with the linear interpolation of coordinates at neighboring high-confidence timepoints.

Immunohistochemistry

Following completion of experiments, rats were deeply anesthetized with isoflurane and transcardially perfused with 0.9% NaCl followed by 10% buffered formalin solution (HT501320, Sigma Aldrich). Brains were removed and stored in formalin and switched to 20% sucrose the following morning. All brains were sectioned at 40 μ m on a freezing stage microtome (SM2010R, Leica Biosystems). Sections were collected and processed to fluorescently tag GFP (as an indicator of GCaMP6f or GRAB_DA2h expression) and/or TH via immunohistochemistry. Primary antibodies were incubated at 4°C (washes and other steps at room temperature). Tissues were washed with 1X Potassium Phosphate Buffered Saline (KPBS) six times for ten-minutes, permeabilized in 0.3% Triton-X 100 for 30-minutes and blocked in 2% normal donkey serum for 30-minutes. Sections were incubated in rabbit anti-TH (AB152, Sigma Aldrich) and/or chicken anti-GFP (AB13907, Abcam) antibodies overnight (~18-hours). After six ten-minute 1xKPBS washes, secondary antibody (Cy3 conjugated donkey anti-rabbit and AF488 conjugated donkey anti-chicken; Jackson ImmunoResearch) was applied and sections were incubated for 2-hours at room-temp followed by a single wash. Sections were then mounted onto glass slides, air dried, and coverslipped with Fluoroshield with DAPI (F6057, Sigma Aldrich). Data from rats with GFP expression and fiber placements within the borders of the VTA or NAc (Paxinos and Charles Watson, 2007 [\[1\]](#)) were included in analyses.

Data Analyses

To quantify results from *in vivo* fiber photometry experiments, the mean z-score during the 5s infusion period was measured on each trial and averaged across trials for each session. One- or two-way repeated measures analysis of variance (ANOVA) or paired t-tests were used for statistical comparisons. When group main effects were found with two or more treatments, Tukey's (for one-way ANOVAs) and Sidak's (for two-way ANOVAs) post hoc tests were employed. Linear regression was used to calculate p values, r^2 goodness-of-fit, 95% confidence bands of the best-fit line, and linear equations for average dopamine response vs. average behavioral reactivity or sucrose preference score. A p value of < 0.05 was used to determine statistical significance. These statistical analyses were performed using GraphPad Prism 10.0 Software (GraphPad Software Inc.).

Finally, we performed ROC analysis (Green and Swets, 1966; Cone et al., 2015 [\[2\]](#)) to quantify how discriminable neural responses were to different test conditions via R Studio using the pROC package (package version 1.18.5, RStudio Team (2020), code available on GitHub (Robin et al., 2011 [\[3\]](#))). For each trial, we measured the mean z-score during the 5s intraoral infusion period. The analysis shows the ability of an ideal observer to classify a neural response as driven by sucrose/Unpaired infusion (compared with quinine/Paired). We plotted the rate of hits as a function of false alarm rate across a range of thresholds determined as the mean of the difference between consecutive values of observed z-scores. We then computed the area under the ROC curve (AUC_{ROC}) and used a threshold of $AUC_{ROC} > 0.7$ to indicate that an ideal observer could reliably discriminate between conditions.

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Additional information

Author Contributions

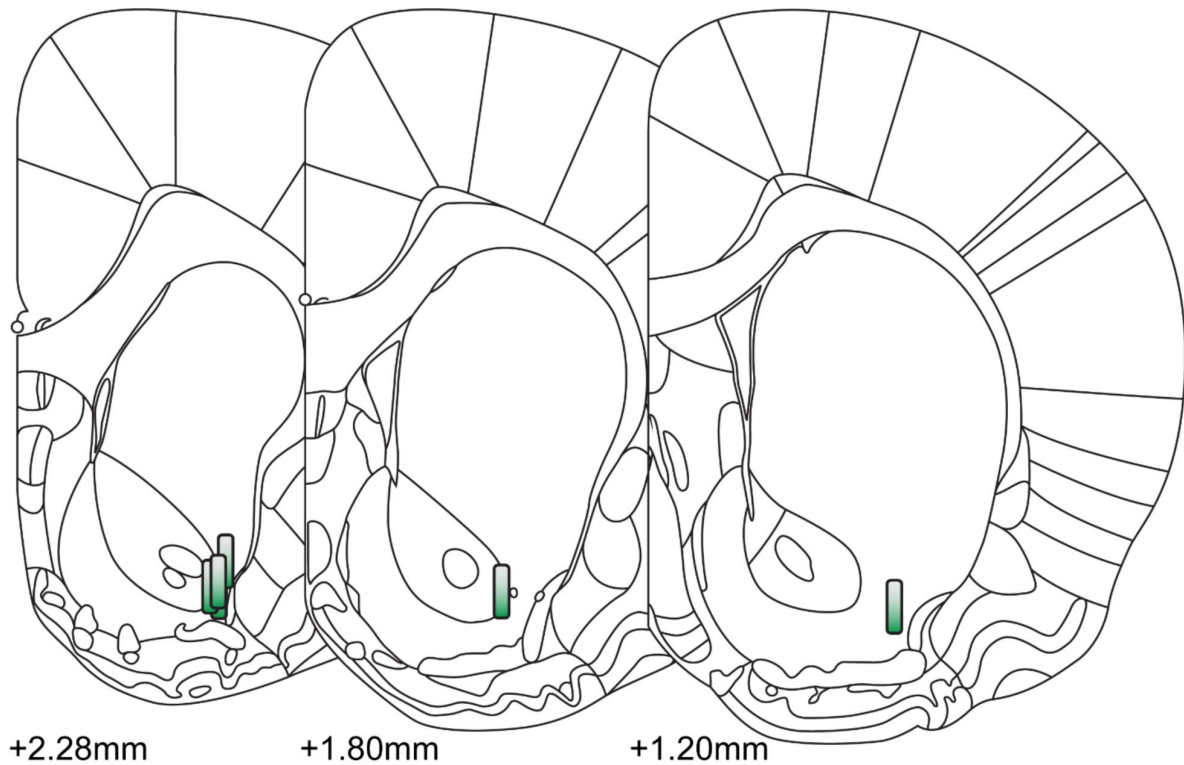
Designed research: MKL, SJH, MFR; Performed research: SJH, MKL, PB, & RD; Data Analysis and Visualization: MKL, RD, ATK, JDR, MFR; Wrote the paper: MKL & MFR.

Ethics Declarations

The authors declare no competing interests.

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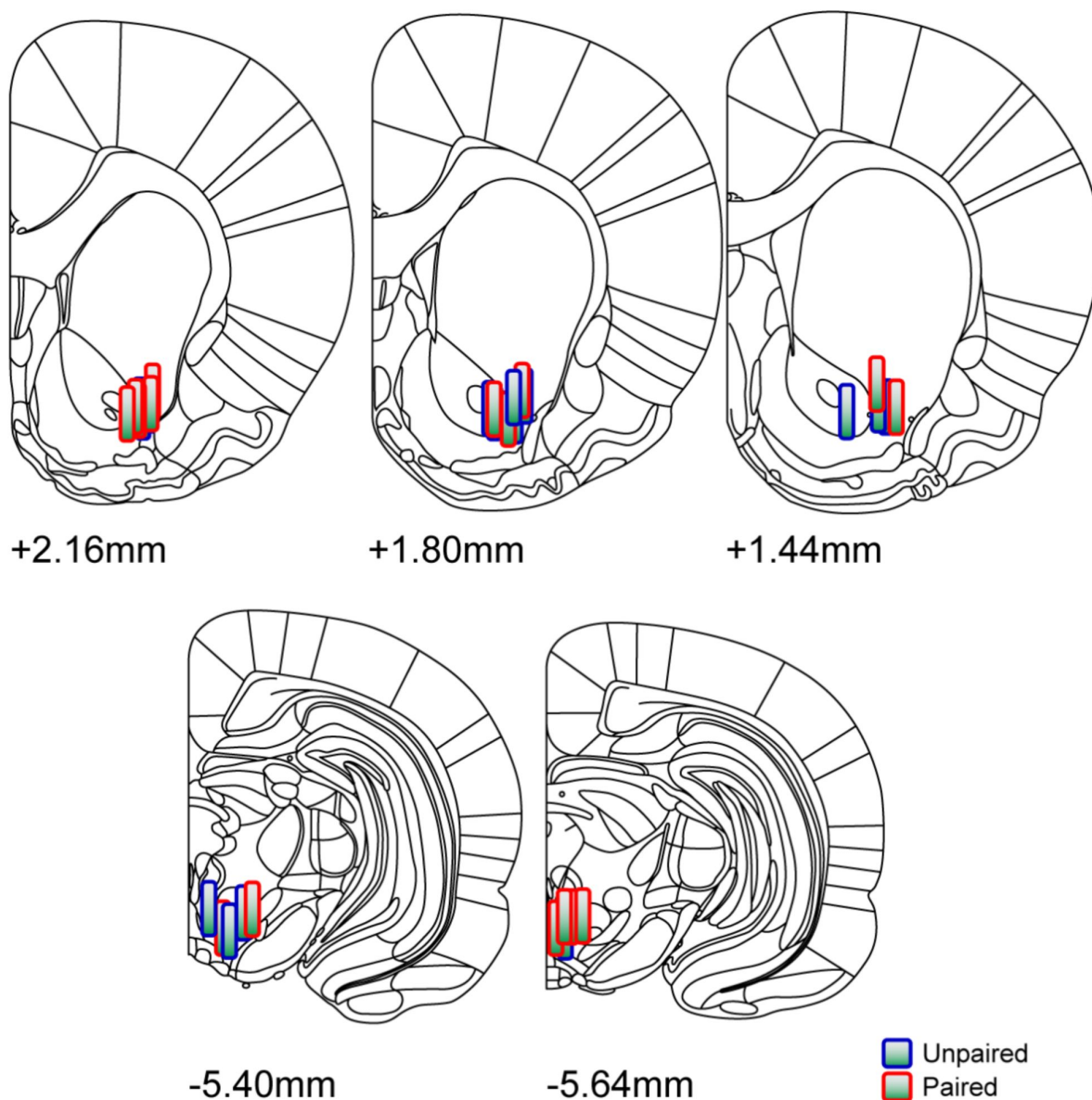
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Supplementary Figure 1.

NAc lateral shell recording placements from Primary Taste experiment.

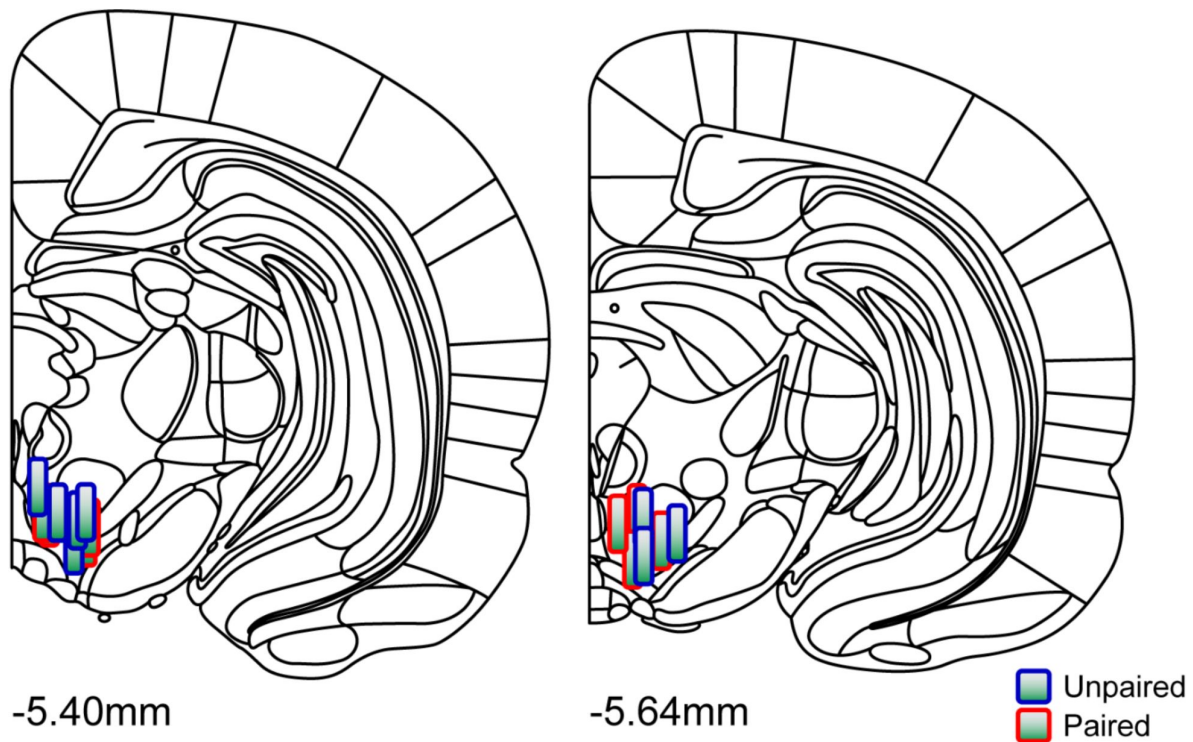
The fiber optic implant locations in the NAc lateral shell were verified via post-experimental histological reconstruction. Measurement values denote anterior-posterior coronal plane in relation to Bregma. Reconstructed images do not represent the specific hemisphere where fiber optics were implanted.



Supplementary Figure 2.

NAc lateral shell and VTA recording placements from Single Pairing CTA experiment.

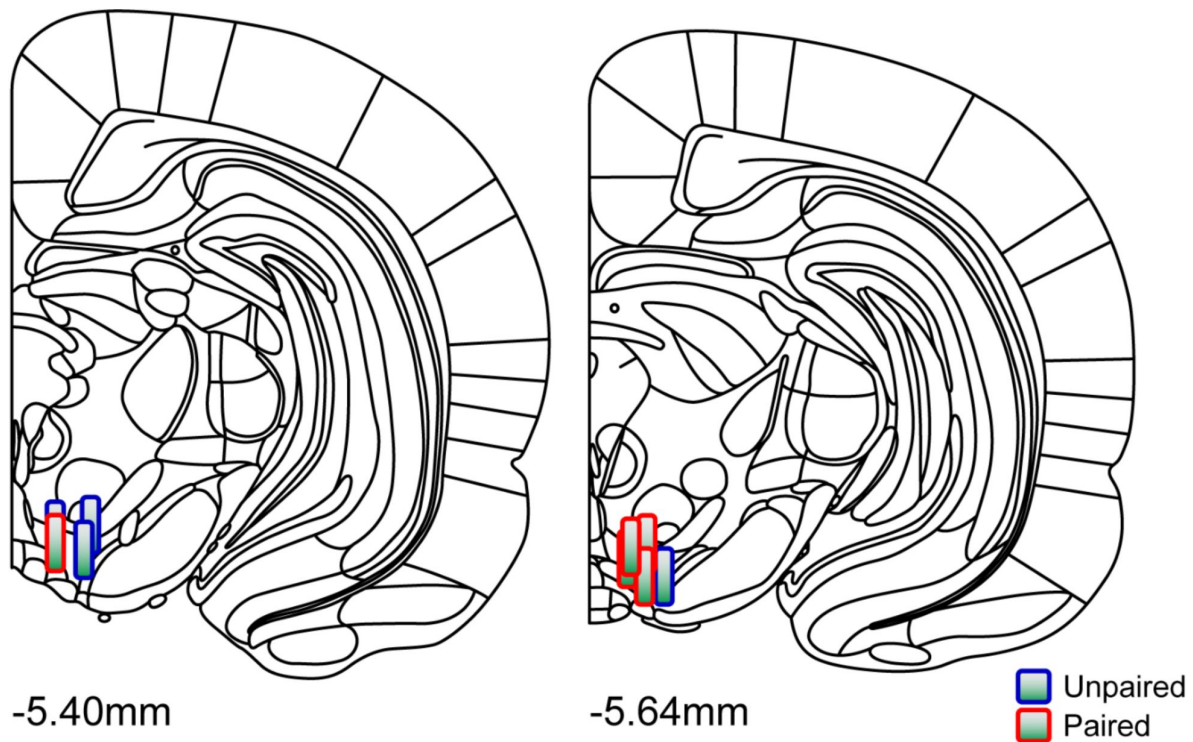
The fiber optic implant locations in the NAc lateral shell and VTA were verified via post-experimental histological reconstruction. Measurement values denote anterior-posterior coronal plane in relation to Bregma. Treatment group is indicated by outline color: Unpaired (blue outline) and Paired (red outline). Reconstructed images do not represent the specific hemisphere where fiber optics were implanted.



Supplementary Figure 3.

VTA recording placements from Delayed Test CTA experiment.

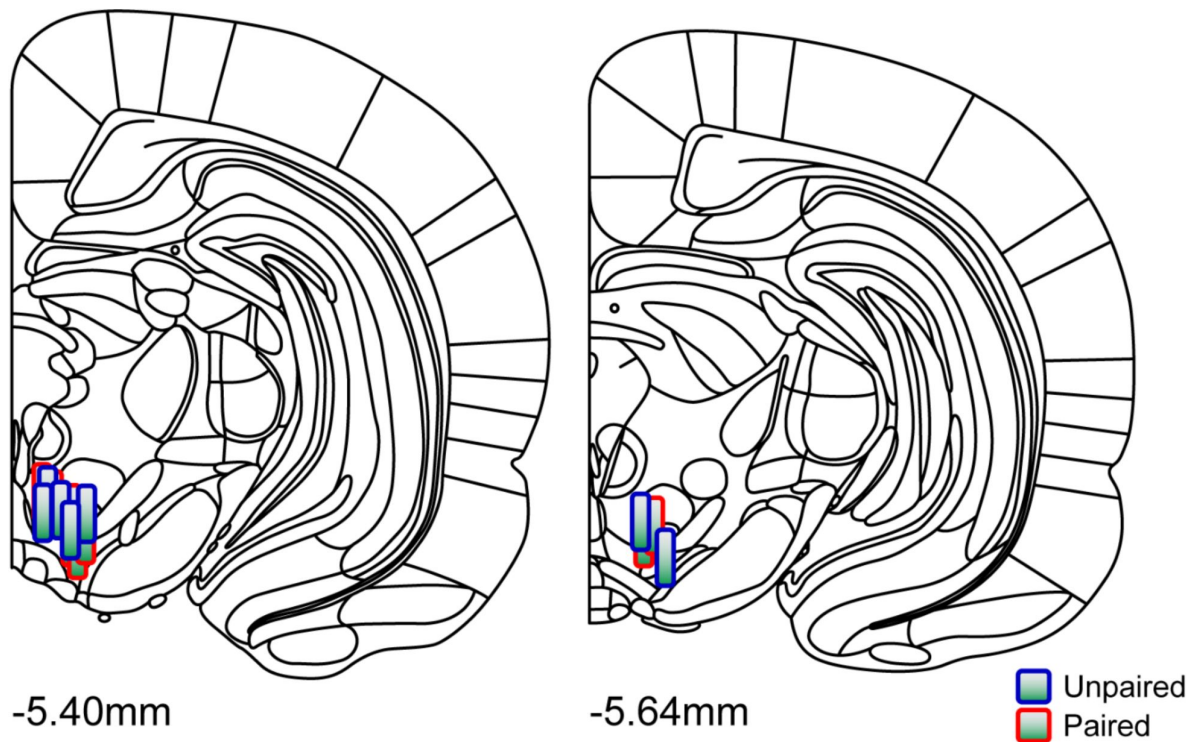
The fiber optic implant locations in the VTA were verified via post-experimental histological reconstruction. Measurement values denote anterior-posterior coronal plane in relation to Bregma. Treatment group is indicated by outline color: Unpaired (blue outline) and Paired (red outline). Reconstructed images do not represent the specific hemisphere where fiber optics were implanted.



Supplementary Figure 4.

VTA recording placements from Single Paring CTA with Extinction experiment.

The fiber optic implant locations in the VTA were verified via post-experimental histological reconstruction. Measurement values denote anterior-posterior coronal plane in relation to Bregma. Treatment group is indicated by outline color: Unpaired (blue outline) and Paired (red outline). Reconstructed images do not represent the specific hemisphere where fiber optics were implanted.



Supplementary Figure 5.

VTA recording placements from Repeated Paring CTA with Extinction experiment.

The fiber optic implant locations in the VTA were verified via post-experimental histological reconstruction. Measurement values denote anterior-posterior coronal plane in relation to Bregma. Treatment group is indicated by outline color: Unpaired (blue outline) and Paired (red outline). Reconstructed images do not represent the specific hemisphere where fiber optics were implanted.

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- Zweifel LS, Fadok JP, Argilli E, Garelick MG, Jones GL, Dickerson TMK, Allen JM, Mizumori SJY, Bonci A, Palmiter RD (2011) **Activation of dopamine neurons is critical for aversive conditioning and prevention of generalized anxiety** *Nat Neurosci* **14** <https://doi.org/10.1038/nn.2808>

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Reviewer #1 (Public review):

Summary:

Loh and colleagues investigate valence encoding in the mesolimbic dopamine system. Using an elegant approach, they show that sucrose, which normally evokes strong dopamine neuron activity and release in the nucleus accumbens, is made aversive via conditioned taste aversion, the same sucrose stimulus later evokes much less dopamine neuron activity and release. Thus, dopamine activity can dynamically track the changing valence of an unconditioned stimulus. These results are important for helping clarify valence and value related questions that are the matter of ongoing debate regarding dopamine functions in the field.

Strengths:

This is an elegant way to ask this question, the within subject's design and the continuity of the stimulus is a strong way to remove a lot of the common confounds that make it difficult to interpret valence-related questions. I think these are valuable studies that help tie up questions in the field while also setting up a number of interesting future directions. There are number of control experiments and tweaks to the design that help eliminate a number of competing hypotheses regarding the results. The data are clearly presented and contextualized.

Weaknesses for consideration:

The focus on one relatively understudied region of the rat striatum for dopamine recordings could potentially limit generalization of the findings. While this can be determined in future studies, the implications should be further discussed in the current manuscript.

<https://doi.org/10.7554/eLife.103260.1.sa3>

Reviewer #2 (Public review):

Summary:

Koh et al. report an interesting manuscript studying dopamine binding in the lateral accumbens shell of rats across the course of conditioned taste aversion. The question being asked here is how does the dopamine system respond to aversion? The authors take advantage of unique properties of taste aversion learning (notably, within-subjects remapping of valence to the same physical stimulus) to address this.

They combine a well controlled behavioural design (including key, unpaired controls) with fibre photometry of dopamine binding via GrabDA and of dopamine neuron activity by gCaMP, careful analyses of behaviour (e.g., head movements; home cage ingestion), the authors show that, 1) conditioned taste aversion of sucrose suppresses the activity of VTA dopamine neurons and lateral shell dopamine binding to subsequent presentations of the sucrose tastant; 2) this pattern of activity was similar to the innately aversive tastant quinine; 3) dopamine responses were negatively correlated with behavioural (inferred taste reactivity) reactivity; and 4) dopamine responses tracked the contingency of between sucrose and illness because these responses recovered across extinction of the conditioned taste aversion.

Strengths:

There are important strengths here. The use of a well-controlled design, the measurement of both dopamine binding and VTA dopamine neuron activity, the inclusion of an extinction manipulation; and the thorough reporting of the data. I was not especially surprised by these results, but these data are a potentially important piece of the dopamine puzzle (e.g., as the authors note, salience-based argument struggles to explain these data).

Weaknesses for consideration:

(1) The focus here is on the lateral shell. This is a poorly investigated region in the context of the questions being asked here. Indeed, I suspect many readers might expect a focus on the medial shell. So, I think this focus is important. But, I think it does warrant greater attention in both the introduction and discussion. We do know from past work that there can be extensive compartmentalisation of dopamine responses to appetitive and aversive events and many of the inconsistent findings in the literature can be reconciled by careful examination of where dopamine is assessed. I do think readers would benefit from

acknowledgement this - for example it is entirely reasonable to suppose that the findings here may be specific to the lateral shell.

(2) Relatedly, I think readers would benefit from an explicit rationale for studying the lateral shell as well as consideration of this in the discussion. We know that there are anatomical (PMID: 17574681), functional (PMID: 10357457), and cellular (PMID: 7906426) differences between the lateral shell and the rest of the ventral striatum. Critically, we know that profiles of dopamine binding during ingestive behaviours there can be highly dissimilar to the rest of ventral striatum (PMID: 32669355). I do think these points are worth considering.

(3) I found the data to be very thoughtfully analysed. But in places I was somewhat unsure:

(a) Please indicate clearly in the text when photometry data show averages across trials versus when they show averages across animals.

(b) I did struggle with the correlation analyses, for two reasons.

(i) First, the key finding here is that the dopamine response to intraoral sucrose is suppressed by taste aversion. So, this will significantly restrict the range of dopamine transients, making interpretation of the correlations difficult.

(ii) Second, the authors report correlations by combining data across groups/conditions. I understand why the authors have done this, but it does risk obscuring differences between the groups. So, my question is: what happens to this trend when the correlations are computed separately for each group? I suspect other readers will share the same question. I think reporting these separate correlations would be very helpful for the field - regardless of the outcome.

(4) Figure 1A is not as helpful as it might be. I do think readers would expect a more precise reporting of GCaMP expression in TH+ and TH- neurons. I also note that many of the nuances in terms of compartmentalisation of dopamine signalling discussed above apply to ventral tegmental area dopamine neurons (e.g. medial v lateral) and this is worth acknowledging when interpreting.

<https://doi.org/10.7554/eLife.103260.1.sa2>

Reviewer #3 (Public review):

Summary:

This study helps to clarify the mixed literature on dopamine responses to aversive stimuli. While it is well accepted that dopamine in the ventral striatum increases in response to various rewarding and appetitive stimuli, aversive stimuli have been shown to evoke phasic increases or decreasing depending on the exact aversive stimuli, behavioral paradigm, and/or dopamine recording method and location examined. Here the authors use a well-designed set of experiments to show differential responses to an appetitive primary reward (sucrose) that later becomes a conditioned aversive stimulus (sucrose previously paired with lithium chloride in a conditioned taste aversion paradigm). The results are interesting and add valuable data to the question of how the mesolimbic dopamine system encodes aversive stimuli, however, the conclusions are strongly stated given that the current data do not necessarily align with prior conflicting data in terms of recording location, and it is not clear exactly how to interpret the generally biphasic dopamine response to the CTA-sucrose which also evolves over exposures within a single session.

Strengths:

- The authors nicely demonstrate that their two aversive stimuli examined, quinine and sucrose following CTA, evoked aversive facial expressions and paw movements that differed

from those following rewarding sucrose to support that the stimuli experienced by the rats differ in valence.

- Examined dopamine responses to the exact same sensory stimuli conditioned to have opposing valences, avoiding standard confounds of appetitive and aversive stimuli being sensed by different sensory modalities (i.e., sweet taste vs. electric shock).
- The authors examined multiple measurements of dopamine activity - cell body calcium (GCaMP6f) in midbrain and release in NAc (Grab-DA2h), which is useful as the prior mixed literature on aversive dopamine responses comes from a variety of recording methods.
- Correlations between sucrose preference and dopamine signals demonstrate behavioral relevance of the differential dopamine signals.
- The delayed testing experiment in Figure 7 nicely controls for the effect of time to demonstrate that the "rewarding" dopamine response to sucrose only recovers after multiple extinction sucrose exposures to extinguish the CTA.

Weaknesses for consideration:

- Regional differences in dopamine signaling to aversive stimuli are mentioned in the introduction and discussion. For instance, the idea that dopamine encodes salience is strongly argued against in the discussion, but the paper cited as arguing for that (Kutlu et al. 2021) is recording from the medial core in mice. Given other papers cited in the text about the regional differences in dopamine signaling in the NAc and from different populations of dopamine neurons in midbrain, it's important to mention this distinction wrt to salience signaling. Relatedly, the text says that the lateral NAc shell was targeted for accumbens recordings, but the histology figure looks like the majority of fibers were in the anterior lateral core of NAc. For the current paper to be a convincing last word on the issue, it would be extremely helpful to have similar recordings done in other parts of the NAc to do a more thorough comparison against other studies.
- Dopamine release in the NAc never dips below baseline for the conditioned sucrose. Is it possible to really consider this as a signal for valence per se, as opposed to it being a weaker response relative to the original sucrose response?
- Related to this, the main measure of the dopamine signal here, "mean z-score," obscures the temporal dynamics of the aversive dopamine response across a trial. This measure is used to claim that sucrose after CTA is "suppressing" dopamine neuron activity and release, which is true relative to the positive valence sucrose response. However, both GRAB-DA and cell-body GCaMP measurements show clear increases after onset of sucrose infusion before dipping back to baseline or slightly below in the average of all example experiments displayed. One could point to these data to argue either that aversive stimuli cause phasic increases in dopamine (due to the initial increase) or decreases (due to the delayed dip below baseline) depending on the measurement window. Some discussion of the dynamics of the response and how it relates to the prior literature would be useful.
 - Would this delayed below-baseline dip be visible with a shorter infusion time?
 - Does the max of the increase or the dip of the decrease better correlate with the behavioral measures of aversion (orofacial, paw movements) or sucrose preference than "mean z-score" measure used here?
 - The authors argue strongly in the discussion against the idea that dopamine is encoding "salience." Could this initial peak (also seen in the first few trials of quinine delivery, fig 1c color plot) be a "salience" response?
- Related to this, the color plots showing individual trials show a reduction in the increases to positive valence sucrose across conditioning day trials and a flip from infusion-onset increase to delayed increases across test day trials. This evolution across days makes it appear that the

last few conditioning day trials would be impossible to discriminate from the first few test day trials in the CTA-paired. Presumably, from strength of CTA as a paradigm, the sucrose is already aversive to the animals at the first trial of test day. Why do the authors think the response evolves across this session?

- Given that most of the work is using a conditioned aversive stimulus, the comparison to a primary aversive tastant quinine is useful. However, the authors saw basically no dopamine response to a primary aversive tastant quinine (measured only with GRAB-DA) and saw less noticeable decreases following CTA for NAc recordings with GRAB-DA2h than with cell body GCaMP. Given that they are using the high-affinity version of the GRAB sensor, this calls into question whether this is a true difference in release vs. soma activity or issue of high affinity release sensor making decreases in dopamine levels more difficult to observe.

<https://doi.org/10.7554/eLife.103260.1.sa1>

Author response:

Reviewer #1 (Public review):

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Loh and colleagues investigate valence encoding in the mesolimbic dopamine system. Using an elegant approach, they show that sucrose, which normally evokes strong dopamine neuron activity and release in the nucleus accumbens, is made aversive via conditioned taste aversion, the same sucrose stimulus later evokes much less dopamine neuron activity and release. Thus, dopamine activity can dynamically track the changing valence of an unconditioned stimulus. These results are important for helping clarify valence and value related questions that are the matter of ongoing debate regarding dopamine functions in the field.

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This is an elegant way to ask this question, the within subject's design and the continuity of the stimulus is a strong way to remove a lot of the common confounds that make it difficult to interpret valence-related questions. I think these are valuable studies that help tie up questions in the field while also setting up a number of interesting future directions. There are number of control experiments and tweaks to the design that help eliminate a number of competing hypotheses regarding the results. The data are clearly presented and contextualized.

Weaknesses for consideration:

The focus on one relatively understudied region of the rat striatum for dopamine recordings could potentially limit generalization of the findings. While this can be determined in future studies, the implications should be further discussed in the current manuscript.

We agree that the manuscript would benefit from providing a stronger rationale for our recording sites and acknowledging the potential for regional differences in dopamine signaling. We have made the following additions to the manuscript:

Added to the Discussion: "Recordings were targeted to the lateral VTA and the corresponding approximate terminal site in the NAc lateral shell (Lammel et al., 2008). Subregional differences in dopamine activity likely contribute to mixed findings on dopamine and affect. For example, dopamine in the NAc lateral shell differentially encodes cues predictive of

rewarding sucrose and aversive footshock, which is distinct from NAc medial shell dopamine responses (de Jong et al., 2019). Our findings are similar to prior work from our group targeting recordings to the NAc dorsomedial shell (Hsu et al., 2020; McCutcheon et al., 2012; Roitman et al., 2008): there, intraoral sucrose increased NAc dopamine release while the response in the same rats to quinine was significantly lower.”

Reviewer #2 (Public review):

Summary:

Koh et al. report an interesting manuscript studying dopamine binding in the lateral accumbens shell of rats across the course of conditioned taste aversion. The question being asked here is how does the dopamine system respond to aversion? The authors take advantage of unique properties of taste aversion learning (notably, within-subjects remapping of valence to the same physical stimulus) to address this.

They combine a well controlled behavioural design (including key, unpaired controls) with fibre photometry of dopamine binding via GrabDA and of dopamine neuron activity by gCaMP, careful analyses of behaviour (e.g., head movements; home cage ingestion), the authors show that, 1) conditioned taste aversion of sucrose suppresses the activity of VTA dopamine neurons and lateral shell dopamine binding to subsequent presentations of the sucrose tastant; 2) this pattern of activity was similar to the innately aversive tastant quinine; 3) dopamine responses were negatively correlated with behavioural (inferred taste reactivity) reactivity; and 4) dopamine responses tracked the contingency of between sucrose and illness because these responses recovered across extinction of the conditioned taste aversion.

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There are important strengths here. The use of a well-controlled design, the measurement of both dopamine binding and VTA dopamine neuron activity, the inclusion of an extinction manipulation; and the thorough reporting of the data. I was not especially surprised by these results, but these data are a potentially important piece of the dopamine puzzle (e.g., as the authors note, salience-based argument struggles to explain these data).

Weaknesses for consideration:

(1) The focus here is on the lateral shell. This is a poorly investigated region in the context of the questions being asked here. Indeed, I suspect many readers might expect a focus on the medial shell. So, I think this focus is important. But, I think it does warrant greater attention in both the introduction and discussion. We do know from past work that there can be extensive compartmentalisation of dopamine responses to appetitive and aversive events and many of the inconsistent findings in the literature can be reconciled by careful examination of where dopamine is assessed. I do think readers would benefit from acknowledgement this - for example it is entirely reasonable to suppose that the findings here may be specific to the lateral shell.

As with our response to Reviewer 1, we agree that we should provide further rationale for focusing our recordings on the lateral shell and acknowledge potential differences in dopamine dynamics across NAc subregions. In addition to the changes in the Discussion detailed in our response to Reviewer 1, we have made the following additions to the Introduction:

Added to the Introduction: “NAc lateral shell dopamine differentially encodes cues predictive of rewarding (i.e., sipper spout with sucrose) and aversive stimuli (i.e., footshock), which is

distinct from other subregions (de Jong et al., 2019). It is important to note that other regions of the NAc may serve as hedonic hotspots (e.g. dorsomedial shell; or may more closely align with the signaling of salience (e.g. ventromedial shell; (Yuan et al., 2021)).”

(2) Relatedly, I think readers would benefit from an explicit rationale for studying the lateral shell as well as consideration of this in the discussion. We know that there are anatomical (PMID: 17574681), functional (PMID: 10357457), and cellular (PMID: 7906426) differences between the lateral shell and the rest of the ventral striatum. Critically, we know that profiles of dopamine binding during ingestive behaviours there can be highly dissimilar to the rest of ventral striatum (PMID: 32669355). I do think these points are worth considering.

There are several reasons why dopamine dynamics were recorded in the NAc lateral shell:

(1) Dopamine neurons in more medial aspects of the VTA preferentially target the NAc medial shell and core whereas dopamine neurons in the lateral VTA – our target for VTA DA recordings – project to the lateral shell of the NAc (Lammel et al., 2008). Thus, our goal was to sample NAc release dynamics in areas that receive projections from our cell body recording sites.

(2) Cues predictive of reward availability (i.e., sipper spout with sucrose) and aversive stimuli (i.e., footshock) are differentially encoded by NAc lateral shell dopamine, which is distinct from NAc ventromedial shell dopamine responses (de Jong et al., 2019). These findings suggest a role for NAc lateral shell dopamine in the encoding of a stimulus’s valence, which made the subregion an area of interest for further examination.

(3) With respect to the medial NAc shell specifically, extensive literature had already shown it to be a ‘hedonic hotspot’ (Morales and Berridge, 2020; Yuan et al., 2021) whereas the ventral portion is more mixed with respect to valence (Yuan et al., 2021). We had previously shown that intraoral infusions of primary taste stimuli of opposing valence (i.e., sucrose and quinine) evoke differential responses in dopamine release within the NAc dorsomedial shell (Roitman et al., 2008). We more recently replicated differential dopamine responses from dopamine cell bodies in the lateral VTA (Hsu et al., 2020) and thus endeavored to the possibility of changing dopamine responses in the lateral VTA to the same stimulus as its valence changes. As a result of these choices, measuring dopamine release in the lateral shell was a logical choice. The field would greatly benefit from continued future work surveying the entirety of the VTA DA projection terminus.

We have included these points of justification in the Introduction and Discussion sections.

(3) I found the data to be very thoughtfully analysed. But in places I was somewhat unsure:

(a) Please indicate clearly in the text when photometry data show averages across trials versus when they show averages across animals.

We have now explicitly indicated in the figure legends of Figures 1, 3, 5, 7, and 8:

(1) In heat maps, each row represents the averaged (across rats) response on that trial.

(2) Traces below heat maps represent the response to infusion averaged first across trials for each rat and then across all rats.

(3) Insets represent the average z-score across the infusion period averaged first across all trials for each rat and then across all rats.

(b) I did struggle with the correlation analyses, for two reasons.

(i) First, the key finding here is that the dopamine response to intraoral sucrose is suppressed by taste aversion. So, this will significantly restrict the range of dopamine transients, making interpretation of the correlations difficult.

The overall hypothesis is that the dopamine response would correlate with the valence of a taste stimulus – even and especially when the stimulus remained constant but its valence changed. We inferred valence from the behavioral reactivity to the stimulus – reasoning that an appetitive taste will evoke minimal movement of the nose and paws (presumably because the animals are primarily engaging in small mouth movements associated with ingestion as shown by the seminal work of Grill and Norgren (1978) and the many studies published by the K.C. Berridge group) whereas an aversive taste will evoke significantly more movement as the rats engage in rejection responses (e.g. forelimb flails, chin rubs, etc.). When we conducted our regression analyses we endeavored to be as transparent as possible and labeled each symbol based on group (Unpaired vs Paired) and day (Conditioning vs Test). Both behavioral reactivity and dopamine responses change – but only for the Paired rats across days. In this sense, we believe the interpretation is clear. However, the Reviewer raises an important criticism that there would essentially be a floor effect with dopamine responses. We believe this is mitigated by data acquired across extinction and especially in Figure 9B. Here, the observations that dopamine responses fall to near zero but return to pre-conditioning levels in the Paired group with strong correlation between dopamine and behavioral reactivity throughout would hopefully partially allay the Reviewer’s concerns. See Part ii below for further support.

(ii) Second, the authors report correlations by combining data across groups/conditions. I understand why the authors have done this, but it does risk obscuring differences between the groups. So, my question is: what happens to this trend when the correlations are computed separately for each group? I suspect other readers will share the same question. I think reporting these separate correlations would be very helpful for the field -

regardless of the outcome.

To address this concern, we performed separate regression analyses for Paired and Unpaired rats and provide the table below to detail results where data were combined across groups or separated. Expectedly, all analyses in Paired rats indicated a significant inverse relationship between dopamine and behavioral reactivity. Afterall, it is only in this group where behavioral reactivity to the taste stimulus changes as function of conditioning. Perhaps even more striking is that in almost all comparisons, even when restricting the regression analysis to Unpaired rats, we still observed a significant inverse relationship between dopamine and behavioral reactivity in most experiments. We have outlined the separated correlations below (asterisks denote slopes significantly different from 0; * $p < 0.05$; ** $p < 0.01$; *** $p < 0.005$; **** $p < 0.001$):

Author response table 1.

Fig.	Statistic	Combined Conditions	Unpaired	Paired
4C	R ²	0.26	0.10	0.29
	P value	0.0009 (****)	0.2058	0.0123 (*)
	Best Fit Equation	Y = -0.025*X + 0.35	Y = -0.017*X + 0.51	Y = -0.024*X + 0.26
4D	R ²	0.22	0.013	0.31
	P value	0.0026 (***)	0.65	0.0093 (**)
	Best Fit Equation	Y = -0.021*X + 0.33	Y = -0.0047*X + 0.58	Y = -0.025*X + 0.23
4G	R ²	0.48	0.53	0.55
	P value	0.0002(****)	0.016 (*)	0.0025 (***)
	Best Fit Equation	Y = -0.057*X - 0.36	Y = -0.053*X - 0.097	Y = -0.057*X - 0.52
4H	R ²	0.51	0.53	0.54
	P value	<0.0001 (****)	0.017 (*)	0.0026 (****)
	Best Fit Equation	Y = -0.044*X - 0.53	Y = -0.040*X - 0.25	Y = -0.044*X - 0.65
6B	R ²	0.38	0.048	0.46
	P value	<0.0001 (****)	0.21	<0.0001 (****)
	Best Fit Equation	Y = 0.33*X + 0.61	Y = 0.037*X + 0.86	Y = 0.46*X + 0.53
9B	R ²	0.43	8.4e-005	0.33
	P value	<0.0001 (****)	0.95	<0.0001 (****)
	Best Fit Equation	Y = -0.045*X + 0.0036	Y = 0.00063*X + 0.71	Y = -0.033*X - 0.11
9C	R ²	0.59	0.15	0.36
	P value	<0.0001 (****)	0.0034 (***)	<0.0001 (****)
	Best Fit Equation	Y = 0.48*X + 0.53	Y = 0.088*X + 0.86	Y = 0.44*X + 0.48

(4) Figure 1A is not as helpful as it might be. I do think readers would expect a more precise reporting of GCaMP expression in TH+ and TH- neurons. I also note that many of the nuances in terms of compartmentalisation of dopamine signalling discussed above apply to ventral tegmental area dopamine neurons (e.g. medial v lateral) and this is worth acknowledging when interpreting t

Others have reported (Choi et al., 2020) and quantified (Hsu et al., 2020) GCaMP6f expression in TH+ neurons. While we didn't report these quantifications, our observations were very much in line with previous quantifications from our laboratory (Hsu et al. 2020).

We agree that we should elaborate on VTA subregional differences and have answered this response above (See responses to Reviewer 1 Weakness #1 and Reviewer 2 Weakness #2).

Reviewer #3 (Public review):

Summary:

This study helps to clarify the mixed literature on dopamine responses to aversive stimuli. While it is well accepted that dopamine in the ventral striatum increases in response to various rewarding and appetitive stimuli, aversive stimuli have been shown to evoke

phasic increases or decreasing depending on the exact aversive stimuli, behavioral paradigm, and/or dopamine recording method and location examined. Here the authors use a well-designed set of experiments to show differential responses to an appetitive primary reward (sucrose) that later becomes a conditioned aversive stimulus (sucrose previously paired with lithium chloride in a conditioned taste aversion paradigm). The results are interesting and add valuable data to the question of how the mesolimbic dopamine system encodes aversive stimuli, however, the conclusions are strongly stated given that the current data do not necessarily align with prior conflicting data in terms of recording location, and it is not clear exactly how to interpret the generally biphasic dopamine response to the CTA-sucrose which also evolves over exposures within a single session.

Strengths:

- The authors nicely demonstrate that their two aversive stimuli examined, quinine and sucrose following CTA, evoked aversive facial expressions and paw movements that differed from those following rewarding sucrose to support that the stimuli experienced by the rats differ in valence.
- Examined dopamine responses to the exact same sensory stimuli conditioned to have opposing valences, avoiding standard confounds of appetitive and aversive stimuli being sensed by different sensory modalities (i.e., sweet taste vs. electric shock)
- The authors examined multiple measurements of dopamine activity - cell body calcium (GCaMP6f) in midbrain and release in NAc (Grab-DA2h), which is useful as the prior mixed literature on aversive dopamine responses comes from a variety of recording methods.
- Correlations between sucrose preference and dopamine signals demonstrate behavioral relevance of the differential dopamine signals.
- The delayed testing experiment in Figure 7 nicely controls for the effect of time to demonstrate that the "rewarding" dopamine response to sucrose only recovers after multiple extinction sucrose exposures to extinguish the CTA.

Weaknesses for consideration:

(1) Regional differences in dopamine signaling to aversive stimuli are mentioned in the introduction and discussion. For instance, the idea that dopamine encodes salience is strongly argued against in the discussion, but the paper cited as arguing for that (Kutlu et al. 2021) is recording from the medial core in mice. Given other papers cited in the text about the regional differences in dopamine signaling in the NAc and from different populations of dopamine neurons in midbrain, it's important to mention this distinction wrt to salience signaling. Relatedly, the text says that the lateral NAc shell was targeted for accumbens recordings, but the histology figure looks like the majority of fibers were in the anterior lateral core of NAc. For the current paper to be a convincing last word on the issue, it would be extremely helpful to have similar recordings done in other parts of the NAc to do a more thorough comparison against other studies.

As the Reviewer notes, NAc dopamine recordings were aimed at the lateral NAc shell. It is possible that some dopamine neurons lying within the anterior lateral core were recorded. Fiber photometry and the size of the fiber optics cannot definitively identify the precise location and number of dopamine neurons from which we recorded. Still, recording sites did not systematically differ between groups. Further, the within-subjects design helps to mitigate any potential biases for one subregion over another. The results presented in the manuscript strongly support a valence code. It is difficult to be the 'last word' on this topic and we suspect debate will continue. We used taste stimuli for appetitive and aversive stimuli

– whereas many in the field will continue to use other noxious stimuli (e.g. foot shock) that likely recruit different circuits en route to the VTA. And there may very well be a different regional profile for dopamine signaling with different noxious stimuli. Moreover, we used intraoral infusion to avoid confounds of stimulus avoidance and competing motivations (e.g. food or fluid deprivation). We believe that this is one of the most important and unique features of our report. Recent work supports a role for phasic increases in dopamine in avoidance of noxious stimuli (Jung et al., 2024) and it will be critical for the field to reflect on the differences between avoidance and aversion. Moreover, in ongoing studies we aspire to fully survey dopamine signaling in conditioned taste aversion across the medial-lateral and dorsal-ventral axes of the VTA and NAc.

(2) Dopamine release in the NAc never dips below baseline for the conditioned sucrose. Is it possible to really consider this as a signal for valence per se, as opposed to it being a weaker response relative to the original sucrose response?

Indeed, NAc dopamine release to intraoral quinine nor aversive sucrose doesn't dip below baseline but rather dopamine binding doesn't change from pre-infusion baseline levels. It should be noted that VTA dopamine cell body activity does indeed dip below baseline in response to aversive sucrose. Moreover, using fast-scan cyclic voltammetry, we showed that dopamine release dips below baseline in the NAc dorsomedial shell in response to intraoral quinine (Roitman et al., 2008). The differences across recording sites may reflect regional differences but they may also reflect differences in recording approaches. GrabDA2h, used here, has relatively slow kinetics that may obscure dips below baseline (see response Weakness# 8 below).

(3) Related to this, the main measure of the dopamine signal here, "mean z-score," obscures the temporal dynamics of the aversive dopamine response across a trial. This measure is used to claim that sucrose after CTA is "suppressing" dopamine neuron activity and release, which is true relative to the positive valence sucrose response. However, both GRAB-DA and cell-body GCaMP measurements show clear increases after onset of sucrose infusion before dipping back to baseline or slightly below in the average of all example experiments displayed. One could point to these data to argue either that aversive stimuli cause phasic increases in dopamine (due to the initial increase) or decreases (due to the delayed dip below baseline) depending on the measurement window. Some discussion of the dynamics of the response and how it relates to the prior literature would be useful.

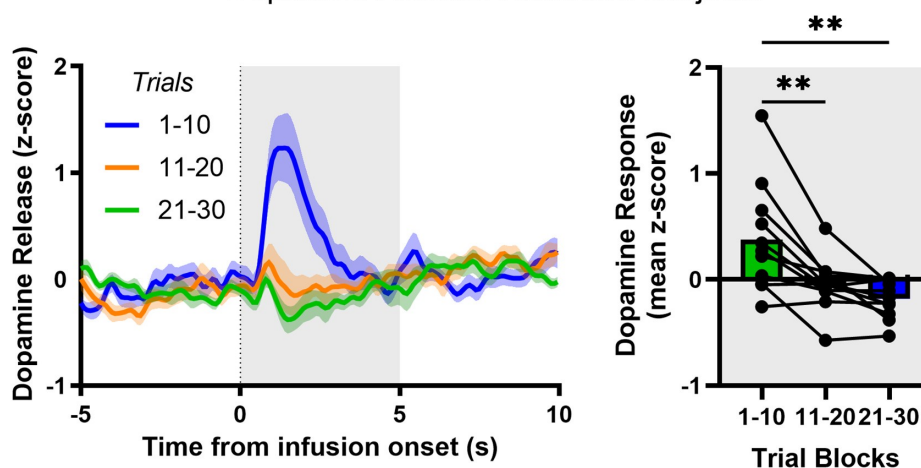
We have used mean z-score to do much of our quantitative analyses but the Reviewer raises the intriguing possibility that we are masking an initial increase in dopamine release and VTA DA activity evoked by aversive taste by doing so. We included the heat maps in the manuscript to be as transparent as possible about the time course of dopamine responses – both within a trial and across trials. The Reviewer's point prompted us to reflect further on the heat maps and recognize that trials early in the session often showed a brief increase in dopamine for aversive sucrose but this response dissipated (NAc dopamine release) or flipped (VTA DA cell body activity) over trials. We now quantitatively characterize this feature by looking at the timecourse of dopamine responses in each third of the trials (1-10, 11-20, 21-30; see Author response images 1,2 and 3). As we infer the valence of the stimulus from nose and paw movements (behavioral reactivity), it is especially striking that we a similar timecourse for changes in behavior. Collectively, the data may reflect an updating process that is relatively slow and requires experience of the stimulus in a new (aversive) state – that is, a model-free process. While our experiments were not designed to test the updating of dopamine responses and discern their participation in model-based versus model-free learning processes – another debate in the dopamine field (Cone et al., 2016; Deserno et al., 2021)– the data reflect a model-free process. This is further supported in the

experiment involving multiple conditioning sessions, where dopamine ‘dips’ are observed in trials 1-10 on Conditioning Day 3 and Extinction Day 1 when the new value of sucrose has been established. Finally, the relatively slow updating of the value of sucrose is reflected in older literature using a continuous intraoral infusion. Using this approach, rats began rejecting the saccharin infusion only after ~2min rather than immediately (Schafe et al., 1998; Schafe and Bernstein, 1996; Wilkins and Bernstein, 2006).

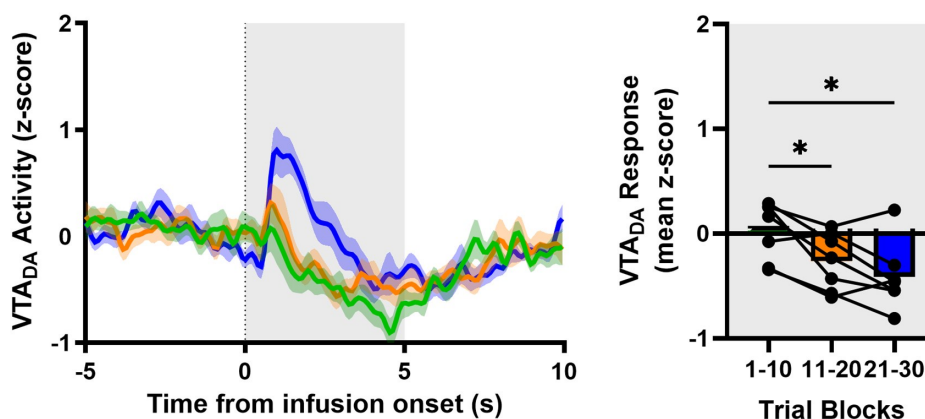
Author response image 1.

Paired Subjects on Test Day

Dopamine Release Recorded Subjects

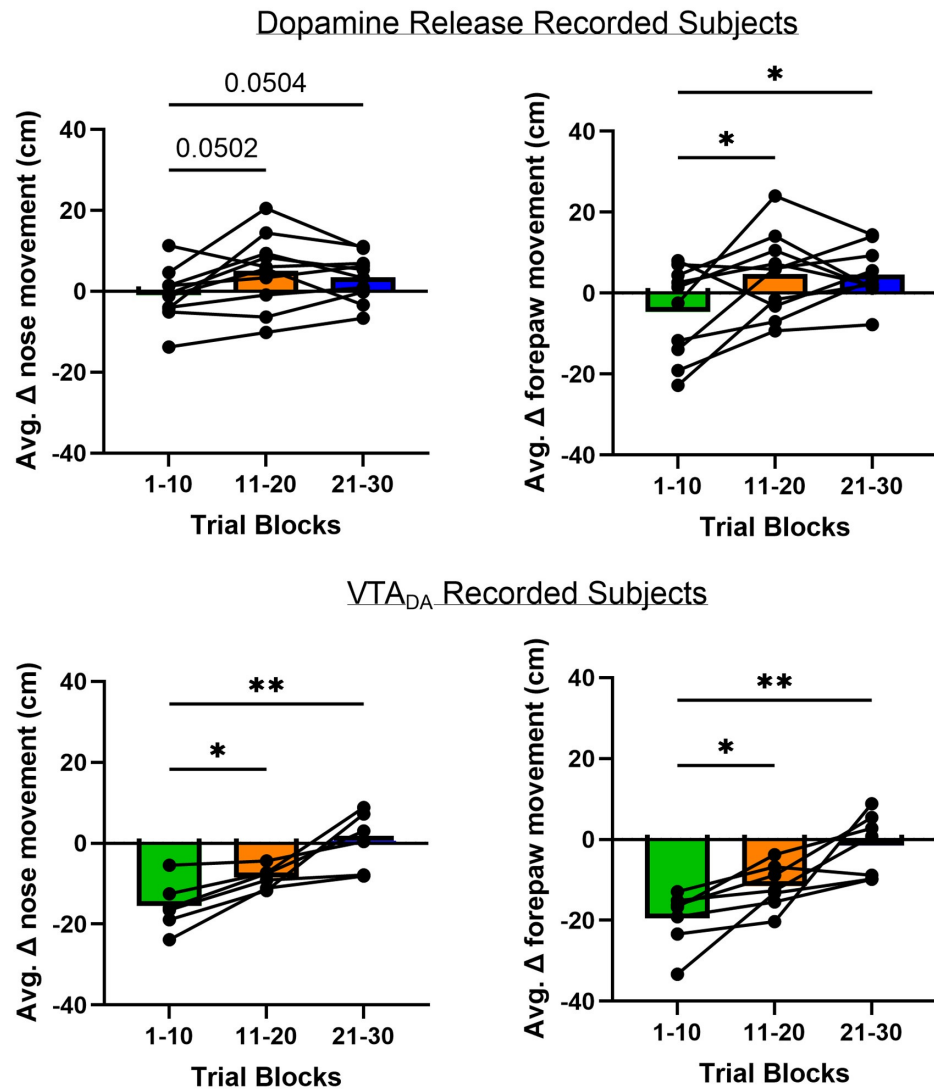


VTA_{DA} Recorded Subjects



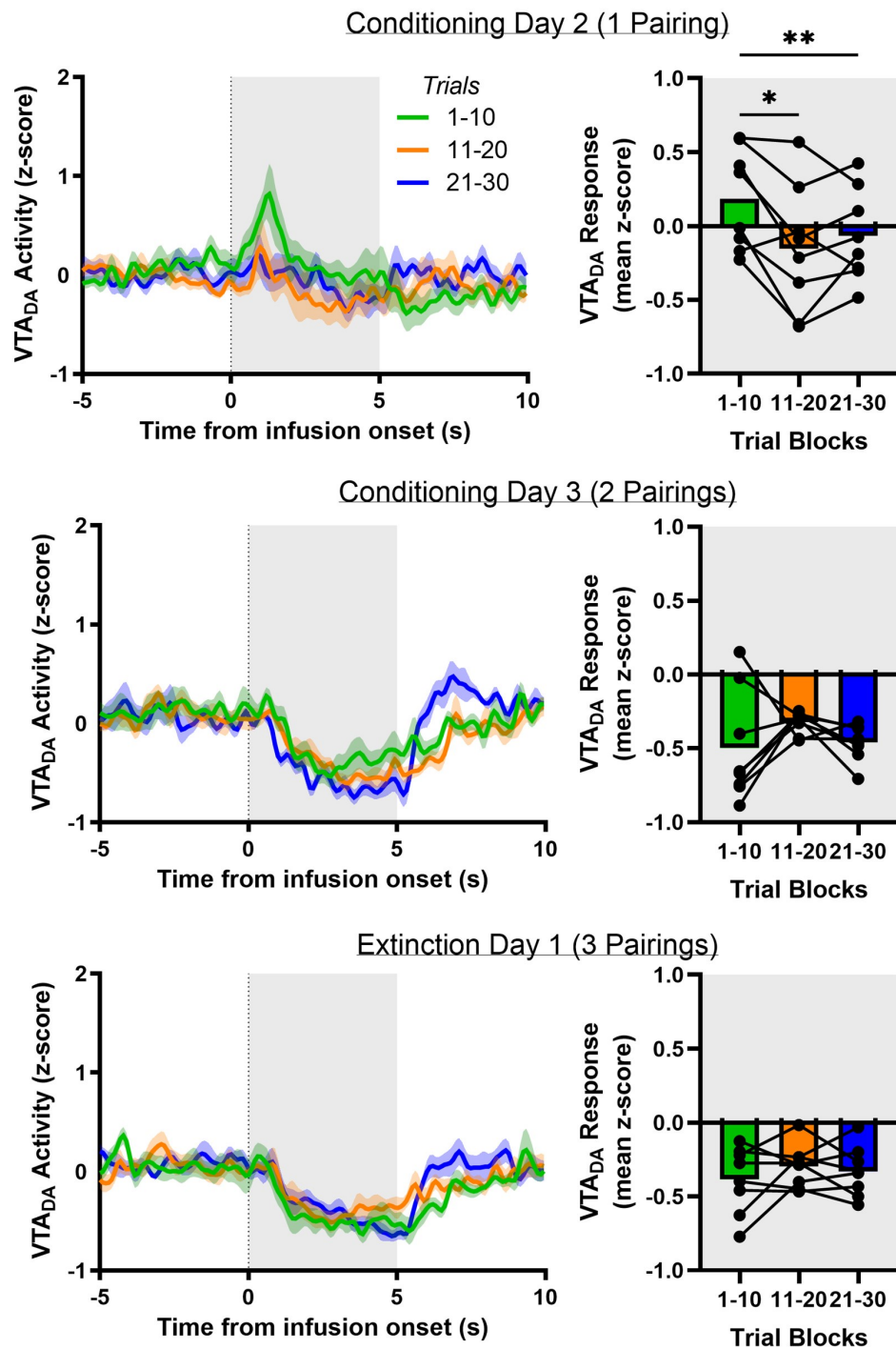
Author response image 2.

Paired Subjects on Test Day



Author response image 3.

Paired Subjects on Test Day



(4) Would this delayed below-baseline dip be visible with a shorter infusion time?

While our experiments did not explore this parameter, it would be interesting to parametrically vary infusion duration times and examine differences in dopamine responses. However, we believe the most parsimonious explanation is that the ‘dip’ in VTA cell body activity develops as a function of the slow updating of the value of sucrose reflective of a model-free process. We recognize that this is mere speculation.

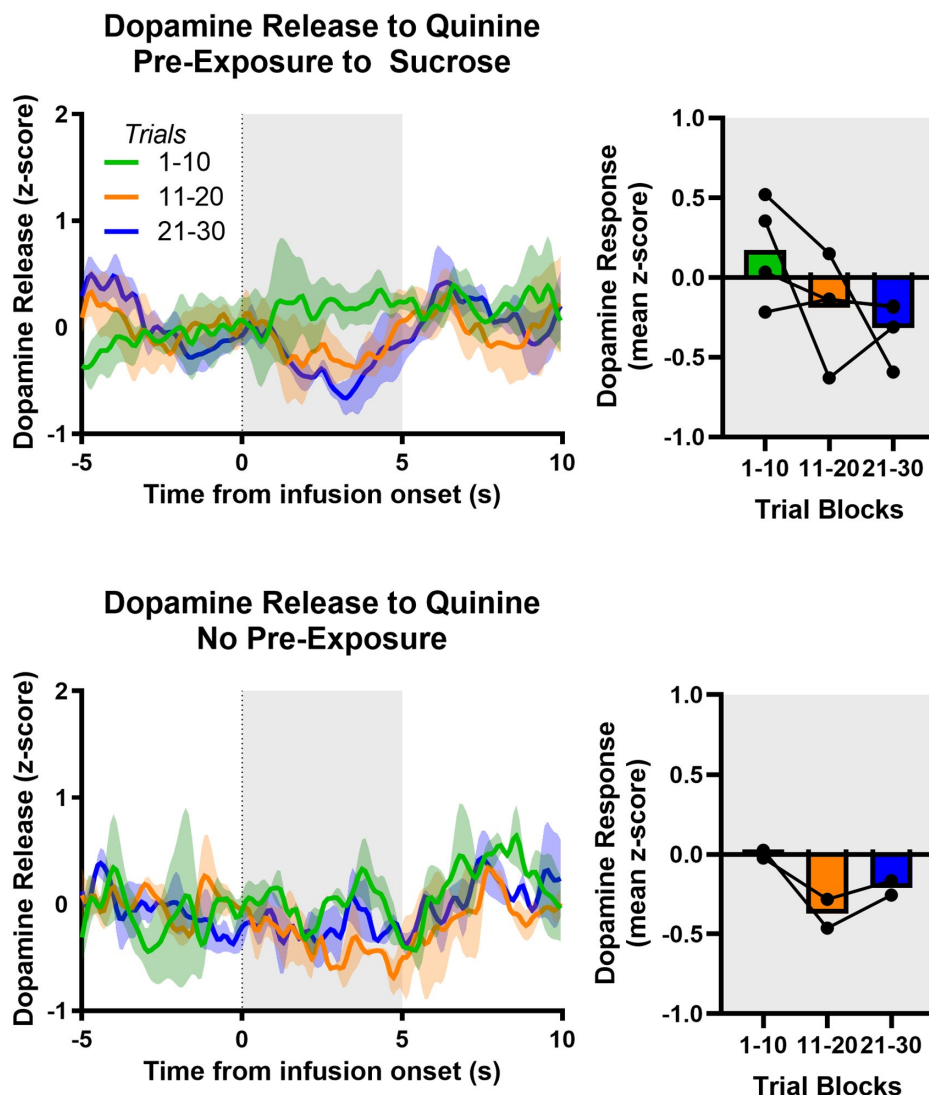
(5) Does the max of the increase or the dip of the decrease better correlate with the behavioral measures of aversion (orofacial, paw movements) or sucrose preference than "mean z-score" measure used here?

It seems plausible that finding the most extreme value from baseline could better correlate to behavioral measures. Time courses to max increase and max decrease are different. Moreover, with appetitive sucrose, there are often multiple transients that occur throughout a single intraoral infusion. Coupled with a noisy time course for individual components of behavioral reactivity, we determined that averaging data across the whole infusion period (i.e. mean z-score) was the most objective way we could analyze the dopamine and behavioral responses to taste stimuli.

(6) The authors argue strongly in the discussion against the idea that dopamine is encoding "salience." Could this initial peak (also seen in the first few trials of quinine delivery, fig 1c color plot) be a "salience" response?

Our response above to the potential for ‘mixed’ dopamine responses to aversive sucrose led to additional analyses that support a slow updating of both behavior and dopamine to the new, aversive value of sucrose. Quinine is innately aversive and thus the Reviewer rightly points out that even here we observe an increase in dopamine release evoked by quinine on the first few trials (as observed in the heat map). We’d like to note, though, that the order of stimulus exposure was counterbalanced across rats. In those rats first receiving a sucrose session, quinine initially caused a modest increase in dopamine release during the first 10 trials (which is more pronounced in the first 2 trials). In the subsequent 2 blocks of 10 trials, no such increase was observed. Interestingly, in rats for which quinine was their first stimulus, we did not see an increase in dopamine release on the first few trials (see Author response image 4). We speculate that the initial sucrose session required the value of intraoral infusions to be updated when quinine was delivered to these rats and that, once more, the updating process may be slow and akin to a model-free process. This analysis, at present, is underpowered but will direct future attention in follow-up work.

Author response image 4.



(7) Related to this, the color plots showing individual trials show a reduction in the increases to positive valence sucrose across conditioning day trials and a flip from infusion-onset increase to delayed increases across test day trials. This evolution across days makes it appear that the last few conditioning day trials would be impossible to discriminate from the first few test day trials in the CTA-paired. Presumably, from strength of CTA as a paradigm, the sucrose is already aversive to the animals at the first trial of test day. Why do the authors think the response evolves across this session?

As the Reviewer noted, Points 3-7 are related. We have speculated that the evolving dopamine response in Paired rats across test day trials reflects a model-free process. Importantly, as in the manuscript, our additional analyses once again show a tight relationship between behavioral reactivity and the dopamine response across the test session trials. It is important to note, though, that these experiments were not designed to test if responses reflect model-free or model-based processes.

(8) Given that most of the work is using a conditioned aversive stimulus, the comparison to a primary aversive tastant quinine is useful. However, the authors saw basically no dopamine response to a primary aversive tastant quinine (measured only with GRAB-DA) and saw less noticeable decreases following CTA for NAc recordings with GRAB-DA2h than with cell body GCaMP. Given that they are using the high-affinity version of the GRAB sensor, this calls into question whether this is a true difference in release vs. soma activity or issue of high affinity release sensor making decreases in dopamine levels more difficult to observe.

We share the same speculation as the Reviewer. Using fast-scan cyclic voltammetry, albeit measuring dopamine concentration in the dorsomedial shell, we observed a clear decrease from baseline with intraoral infusions of quinine (Roitman et al., 2008). Using fiber photometry here, the Reviewer and we note that GRAB_DA2h is a high-affinity (i.e., EC50: 7nM) dopamine sensor with relatively long off-kinetics (i.e., t_{1/2} decay time: 7300ms) (Labouesse et al., 2020). It may therefore be much more difficult to observe decreases (below baseline) using this sensor. The publication of new dopamine sensors - with lower affinity, faster kinetics, and greater dynamic range (Zhuo et al., 2024) – introduces opportunities for comparison and the greater potential for capturing decreases below baseline. Due to the poorer kinetics associated with GRAB_DA2h, we would not assert that direct comparisons between the GCaMP- and GRAB-based signals observed here represent true differences between somatic and terminal activity.

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